

SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

Anonymous authors
Anonymous submission

ABSTRACT

Spatial reasoning is a fundamental aspect of human cognition, yet it remains a major challenge for contemporary vision–language models (VLMs). Prior work largely relied on synthetic or LLM-generated environments with limited task designs and puzzle-like setups, failing to capture the real-world complexity, visual noise, and diverse spatial relationships that VLMs encounter. To address this, we introduce **SPATIALAB**, a comprehensive benchmark for evaluating VLMs’ spatial reasoning in realistic, unconstrained contexts. SPATIALAB comprises 1,400 visual question–answer pairs across six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each with five subcategories, yielding 30 distinct task types. Each subcategory contains at least 25 questions, and each main category includes at least 200 questions, supporting both multiple-choice and open-ended evaluation. Evaluating a broad and directly comparable panel of state-of-the-art VLMs—spanning open- and closed-source, reasoning-focused, and spatially specialized models—against a human baseline exposes a wide and remarkably consistent gap in spatial reasoning. Under the multiple-choice protocol, the strongest model, InternVL3.5-72B, reaches only 54.93% accuracy, trailing the human baseline of 87.57% by 32.64 percentage points. The open-ended protocol widens the gap further: every model loses roughly 10–25 points relative to its own multiple-choice score, and the best system, GPT-5-mini, tops out at 40.93% against a human baseline of 64.93%, a 24-point shortfall. Across both protocols, the ordering of models is stable and the human margin is uniform, pinpointing where the deficits concentrate: complex spatial relationships, depth perception, navigation, and 3D geometry. By providing a diverse, real-world evaluation framework with a common human reference point, SPATIALAB quantifies these critical challenges and the opportunities they open for advancing VLMs’ spatial reasoning, offering a benchmark to guide future research toward robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a fundamental aspect of human cognition that involves understanding spatial layouts, imagining and manipulating objects, and navigating environments. It supports applications ranging from robotics, autonomous driving and augmented/virtual reality to geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents, autonomous systems that perceive and act in dynamic environments, reliable spatial competence requires interpreting geometric relations, depth cues, occlusion patterns, and agent-centric perspectives in scenes that are visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models have advanced multimodal representation and language grounding, but their spatial judgments remain fragile in realistic environments (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks, however, are narrow and simplified, and their reported numbers are correspondingly optimistic. Most focus on elementary tasks such as binary spatial relations, low-resolution depth categories, or schematic navigation cues, often in synthetic or simplified settings that reduce visual clutter and homogenize objects (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024;

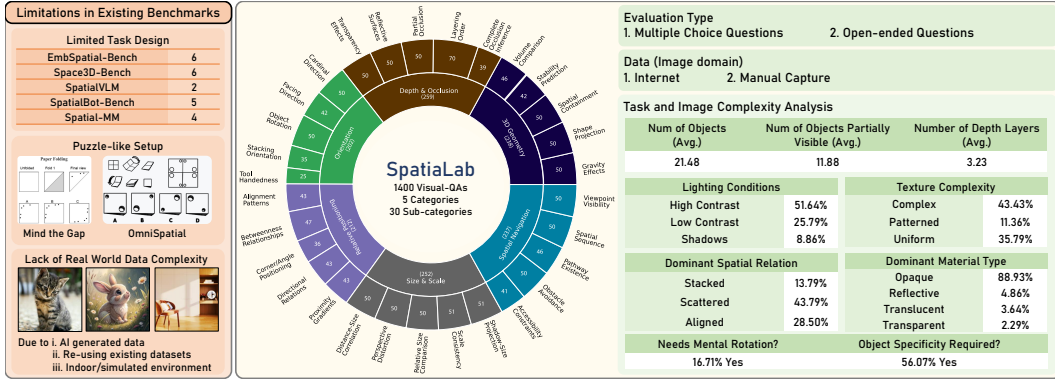


Figure 1: **Overview of SPATIALLAB.** The benchmark addresses limitations of prior datasets (left), introduces 1,400 visual QA pairs spanning 5 categories and 30 subcategories (center), and enables systematic side-by-side evaluation of models against a common human baseline through multiple-choice and open-ended tasks. It features diverse task and image complexity, with varied object counts, layers, lighting, textures, relations, and materials (right).

Qi et al., 2025; Cai et al., 2024). These controlled setups lessen perceptual and reasoning demands, so best-baseline accuracies on them climb high and produce apparent saturation that masks failures under distribution shift. Critical challenges such as robust occlusion inference, scale-consistent judgments across viewpoints, and planning under partial observability remain under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). As a result, models that appear strong on existing benchmarks drop sharply on integrated, multi-step inference tasks, limiting safe deployment in real-world environments where spatial errors carry practical consequences.

A meaningful evaluation must therefore span the all core axes of spatial reasoning: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Unlike current task-isolated setups, human cognition integrates these dimensions seamlessly under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). Robust benchmarks should therefore test both choice and generation, benchmark against human performance, and draw from real-world scenes that reveal failures masked by simulator datasets—principles that motivate our benchmark.

To address the above needs, we introduce **SPATIALLAB**, a benchmark designed to assess spatial reasoning of vision–language models in realistic, unconstrained visual contexts, as outlined in Figure 1. **SPATIALLAB** provides a balanced suite of visual question–answer items covering a broad taxonomy of spatial tasks and supports both multiple-choice and open-ended evaluation formats. The benchmark emphasizes photographic diversity, complex spatial relational queries, and tasks that require integrating depth, perspective, and 3D structure. We evaluate a wide collection of state-of-the-art VLMs, including open- and closed-source models and reasoning-focused systems, compare them to human performance, and perform extensive error analysis. Our key contributions are as follows:

1. We introduce **SPATIALLAB**, a large-scale benchmark of 1,400 visual question–answer pairs spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it ensures balanced coverage and supports both multiple-choice (SPATIALLAB-MCQ) and open-ended (SPATIALLAB-OPEN) evaluation.
2. We perform a **comprehensive, directly comparable evaluation** of 25+ state-of-the-art vision–language models: open- and closed-source, reasoning-oriented, and spatially specialized reasoning models, all scored under identical conditions against a shared human baseline. The comparison exposes a large and consistent gap—over 30 points in multiple-choice and roughly 24 points in open-ended accuracy—that persists across every model family.
3. We deliver **in-depth error analyses and diagnostic studies** that uncover systematic failure modes in depth perception, navigation, occlusion, and 3D geometry, yielding concrete insights and actionable directions for advancing robust, human-aligned spatial understanding.

Table 1: Head-to-head comparison of spatial reasoning benchmarks. Reading across the rows, SPATIALAB offers the broadest taxonomy (30 task categories vs. 2–14 for prior work) at the highest difficulty and complexity, and the lowest best-baseline accuracy (54.93% MCQ / 40.93% open-ended vs. the 45–72% typical of earlier suites), indicating substantially more unsolved headroom.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval		A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth				
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M.E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93

4. We explore multiple strategies to enhance spatial reasoning in VLMs, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, analyzing their benefits and limitations across multiple-choice and open-ended tasks.

The quantitative picture is consistent and striking. In the multiple-choice setting, even the strongest models cluster at roughly 50–55% accuracy while human annotators exceed 85%, a gap of more than 30 percentage points that no model family closes. Open-ended generation is uniformly harder: every model loses another 10–25 points relative to its own multiple-choice score, leaving the best systems near 40% against about 65% for humans. These margins hold across proprietary, open-source, reasoning-oriented, and spatially specialized models, so the human advantage is a property of the task rather than of any single architecture. Error analysis localizes the shortfall to occlusion inference, scale-consistency, and multi-step navigational planning, indicating limits in compositional spatial representations and reasoning. Our comprehensive study further quantifies which interventions deliver robust gains, which amplify existing biases, and how large the practical effect of perceptual grounding is for generalizable spatial reasoning. By providing tasks, directly comparable baseline results against a shared human reference, and diagnostic analyses, SPATIALAB establishes a foundation for measurable progress and reproducible comparison in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning, the ability to perceive, represent, and manipulate spatial relations among objects, plays a central role in human cognition and in enabling embodied agents to navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, we can describe a scene as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes (e.g., position, orientation, size) in a world coordinate system W . Visual-spatial reasoning then seeks to learn a mapping $f : (I, Q) \mapsto R$, where I is an image (or image sequence), Q is a linguistic query, and R is a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ capturing predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), learning f is highly challenging because it requires integrating noisy visual perception with compositional linguistic abstraction under embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Early approaches to spatial reasoning in VLMs have largely relied on basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), often situated in synthetic datasets like CLEVR (Johnson et al., 2016) or structured question-answering benchmarks such as GQA (Hudson & Manning, 2019) or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025). Table 1 makes the contrast concrete: prior suites top out at 2–14 task categories and report best-baseline accuracies of roughly 45–72%, whereas SPATIALAB spans 30 task types at high-to-very-high difficulty and drives the best baseline down to 54.93% (MCQ) and 40.93% (open-ended), leaving far more measurable headroom¹. While useful for

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

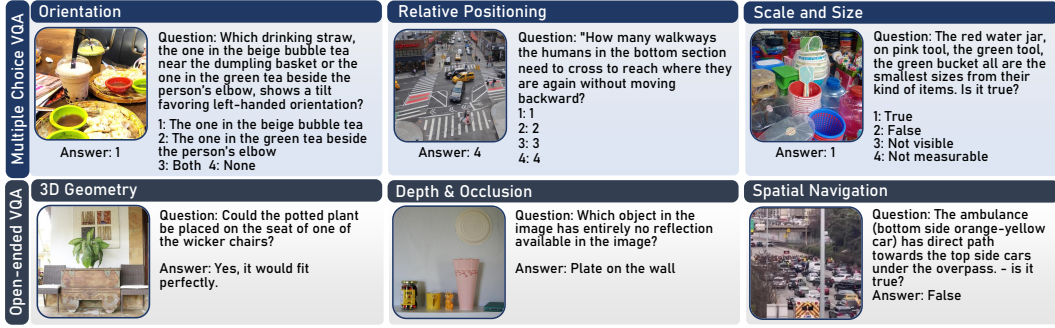


Figure 2: Representative examples from the six categories in both open-ended and MCQ tasks, illustrating the realistic clutter and multi-cue reasoning that drive the large human–model accuracy gaps we report.

controlled evaluation, these settings oversimplify spatial structure, omit environmental noise, and neglect dynamic factors such as temporal changes or viewpoint shifts. To overcome these limitations, we argue for the need to benchmark spatial reasoning in rich, real-world contexts that capture scene complexity, temporal variation, and multiple perspectives. Such settings expose VLMs to challenges they would encounter in embodied or interactive applications, including robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

In SPATIALAB, we organize spatial reasoning into six principal categories, *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each linked to a distinct cognitive faculty studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is further decomposed into five subcategories, producing a taxonomy of thirty task types that isolate fine-grained reasoning skills, such as directional alignment, occlusion inference, or stability prediction. This classification scheme is motivated by the need to capture the full spectrum of visual–spatial reasoning challenges. By systematically breaking down complex reasoning into measurable and testable units, we can evaluate not only whether models “solve” spatial tasks but also which specific dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

Moreover, we adopt two complementary evaluation formats: multiple-choice and open-ended question answering. The multiple-choice format probes recognition and fine discrimination among distractors, while the open-ended format measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Reporting both on the same items lets us quantify the format effect directly—an average 23-point drop from multiple-choice to open-ended across models—so that a single leaderboard number cannot overstate a model’s practical competence. By combining a principled taxonomy, diverse real-world imagery, and dual evaluation formats scored against a common human baseline, SPATIALAB provides a structured framework for systematic benchmarking. This setup enables fine-grained, apples-to-apples model comparison, precise measurement of the residual gap to human-level performance, and deeper insights into the development of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains a persistent challenge for vision–language models (VLMs), especially in naturalistic conditions where occlusion, perspective distortion, and noisy backgrounds co-occur. To address this, we present SPATIALAB, a benchmark designed to systematically evaluate VLMs’ capacity for complex spatial cognition under realistic visual conditions. The benchmark comprises 1,400 visual question–answer pairs organized into six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*, each further divided into five subcategories, yielding 30 distinct task types. Every subcategory includes at least 25 QA pairs, while each top-level category contributes a minimum of 200, ensuring balanced coverage across the taxonomy. SPATIALAB supports both multiple-choice and open-ended evaluation, enabling assessment of discriminative accuracy as well as generative reasoning and explanatory competence,

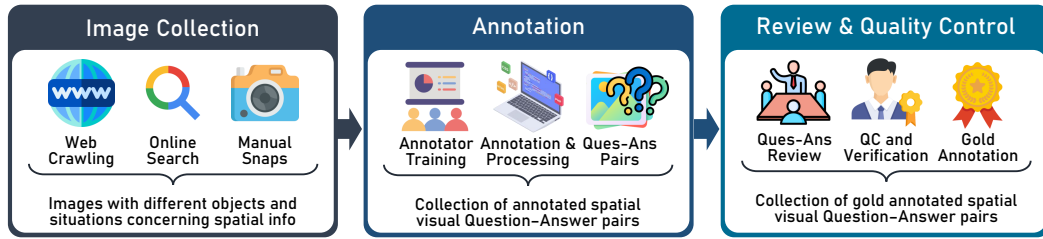


Figure 3: **Data creation pipeline for SPATIALAB.** Images are collected via web crawling, targeted search, and manual snapshots, followed by structured annotation of spatial question–answer pairs.

thus aligning with broader multimodal evaluation practices while emphasizing spatial-specific reasoning. High visual diversity, fine-grained annotation, and rigorous quality control make **SPATIALAB** a reliable and scalable testbed, and Figure 2 illustrates representative examples from the six categories, showcasing both open-ended and multiple-choice formats that capture the richness and difficulty of real-world spatial reasoning tasks.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was curated using three complementary strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manual snapshots collected in diverse indoor and outdoor environments (Figure 3). This multi-source pipeline ensures diversity across object categories, environmental conditions, and spatial configurations. To explicitly capture natural variability, images were profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Each dimension was systematically represented in the final corpus, ensuring that the benchmark reflects a broad spectrum of real-world, in-the-wild scenarios rather than simplified or synthetic setups. Complexity analysis of the corpus quantifies just how far these scenes depart from synthetic setups: images contain 21.48 objects on average, of which 11.88 (55%) are only partially visible, and they are distributed across 3.23 distinct depth layers. Spatial reference chains (e.g., “object A is left of B, which is behind C”) average 2.07 links, so a typical question demands genuinely multi-step relational reasoning rather than a single lookup. In addition, 16.71% of questions require mental rotation and 56.07%—a clear majority—require object specificity (a specific color, pattern, or other distinguishing detail). Taken together, these statistics confirm that **SPATIALAB** systematically captures the cluttered, layered, and relationally rich scenes that characterize real-world spatial reasoning yet remain underrepresented in synthetic datasets.

Annotation. Annotation proceeded in three phases. First, annotators underwent targeted training (details are available in Appendix C.2) to ensure consistency in interpreting spatial categories and subcategories. Second, each image was paired with one or more spatial QA tasks covering a range of reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction. Questions were designed to balance perceptual grounding with higher-order inference (e.g., predicting stability under gravity constraints). Third, QA pairs were encoded in both multiple-choice (04 options) and open-ended formats, ensuring comparability across evaluation paradigms. In total, it produces around 1,500 spatial visual QA pairs, across six categories and 30 subcategories.

Review and Quality Control. To ensure reliability, annotation outputs underwent a three-tier review and validation pipeline. In the first stage, QA items were checked for semantic validity and alignment with the designated subcategory. In the second stage, independent annotators verified correctness of both the question phrasing and the designated answer, focusing on eliminating ambiguity and enforcing task clarity. In the third stage, a gold-standard annotation round was conducted, establishing a set of high-confidence QA pairs for evaluation. All metadata was removed from the images, and the dataset was carefully curated to avoid copyright concerns, though minor oversights may still occur. This procedure yielded a final benchmark of 1,400 validated QA items.

Table 2: Per-category multiple-choice accuracy (% , $\uparrow$) on SPATIALAB-MCQ, with all models scored against a shared human baseline. The best overall model, InternVL3.5-72B (54.93%), still trails humans (87.57%) by 32.64 points, and every model in every group sits far below the human row—no family closes the gap on any category. Per-group best scores are bolded and underlined; the Overall column is shaded, and column headers list the number of questions per category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

4 EXPERIMENTS AND EVALUATION

Models. We categorize the evaluated models into four groups. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** comprise Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** comprise SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE we only performed open-ended evaluation, as these models are not instruction-tuned and were trained primarily on open-ended tasks, resulting in excessive refusals during MCQ evaluation. For ease of comparison in our results tables, we use color coding: proprietary, open-source up to 7B, larger than 7B, proprietary reasoning, open-source reasoning, spatial specialists, and human baseline.

Evaluation Process and Metrics. We measure accuracy as the primary metric for both MCQ and open-ended tasks. For **MCQ evaluation**, we use direct prompting, where models return the option number corresponding to their selected answer. Responses are automatically checked against the correct answer. Detailed prompts and details are provided in Appendix D.1. For **open-ended evaluation**, models are prompted to generate free-form answers, which are then assessed using a large language model judge. Prompts and details are available Appendix D.2. To validate the open-ended scoring, we quantified how closely the LLM judge tracks humans through an LLM-human agreement

Table 3: Per-category open-ended accuracy (% , $\uparrow$) on SPATIALAB-OPEN, scored against the same human baseline used throughout. Scores are uniformly lower than in the multiple-choice setting: the top model, GPT-5-mini (40.93%), falls 24.00 points short of the human baseline (64.93%), while small open-source models drop as low as 9.64%. Per-group best scores are bolded and underlined; the Overall column is shaded, and column headers list the number of questions per category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Proprietary Models</i>							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
<i>Open-Source Models</i>							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
<i>Reasoning Models</i>							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
<i>Spatial Reasoning Models</i>							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
<i>Human Baseline</i>	73.53	50.19	70.30	69.81	65.48	62.87	64.93

analysis, as detailed in Appendix D.2. As reported in Table 5, the LLM judge (Gemini-2.5-Flash) agrees with human annotators at a level comparable to the humans’ agreement among themselves: its Cohen’s κ against the majority vote is 0.738 (and 0.681–0.795 against individual annotators), essentially on par with the 0.774 Fleiss’ κ measured across the human annotators. Combined with 0.880 raw accuracy against the majority vote, these figures indicate that the automatic judge is a reliable stand-in for human grading rather than a noisy proxy. For additional details on experiment setup, prompts, and evaluation procedures, see Appendix D.

Improving Visual Reasoning Capabilities. To test how far spatial reasoning on our diverse benchmark can be pushed, we compare five strategies under a common protocol: inherent reasoning in the VLMs (Appendix H.1), Chain-of-Thoughts (CoT) (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT, Qwen-VL2.5-3B-Instruct was fine-tuned on a stratified 40% split and evaluated on the held-out 60%, where it improved multiple-choice accuracy consistently across relational and geometric tasks and reduced errors most in navigation and orientation; combining CoT with SFT produced additive gains. The comparison also draws a sharp boundary: agentic reasoning helps orientation substantially but leaves other categories flat or worse, showing that multi-step reasoning alone cannot substitute for deeper perceptual and spatial grounding.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports multiple-choice accuracy on SPATIALAB-MCQ, and the head-line comparison is unambiguous: overall model accuracies span only about 30–55%—barely above the 25% random-choice floor at the low end—while the human baseline reaches 87.57%, leaving at

least a 32.64-point margin even for the single best model. That best model is an open-source system, InternVL3.5-72B, at 54.93%. A cluster of reasoning- and instruction-tuned models sits just behind it within a narrow 1.7-point band (GPT-5-mini: 54.29%, o4-mini: 53.21%, Gemini-2.5-Flash-Thinking: 52.93%), joined by very large open-source models (InternVL3.5-72B: 54.93%, Llama-3.2-90B: 50.36%). Scale, however, is not decisive: Llama-3.2-11B reaches only 30.50% despite its size, roughly 24 points below the leaders and close to chance. The gap also varies sharply by category. Orientation and 3D Geometry are the relative bright spots—several models clear 60%—whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are both harder and far more variable across models. Tellingly, the spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) land only in the low-to-mid 40s overall (41.36%, 40.64%, 40.14%), trailing generalist leaders by more than 13 points, so specialization does not translate into higher accuracy on our diverse question set. The wide within-category spread on Depth & Occlusion and Size & Scale—some architectures handle occlusion well while others fail—points to differing inductive biases or training data. In sum, instruction-tuning and reasoning capability buy the most on geometric and orientation reasoning, while multi-step grounding tasks such as navigation remain a consistent weakness for every family.

SPATIALAB-OPEN. The open-ended scores in Table 3 are uniformly and substantially lower than their multiple-choice counterparts, and the spread is even wider: overall accuracy runs from 9.64% (InternVL3.5-1B) to 40.93% (GPT-5-mini), against a human baseline of 64.93%—so even the strongest model is 24.00 points short of humans, and the weakest is off by more than 55 points. Proprietary models lead the leaderboard (GPT-5-mini: 40.93%, Gemini-2.5-Flash: 30.93%, GPT-4o-mini: 26.00%), but the sharpest signal is the reasoning-tuned advantage: o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) clear most open-source counterparts by 10–20 points, indicating that instruction/CoT-style tuning substantially improves generative outputs. Small open-source models cluster at the bottom in the teens or below (InternVL3.5-1B: 9.64%, Qwen-VL2.5-3B: 12.93%), and scaling helps only modestly—the largest open models reach just the low-20s (InternVL3.5-72B: 23.36%, Qwen-VL2.5-72B: 24.64%), a gain of only about 12 points for a large increase in parameters. The spatial specialists again lag the field, at just 10.36–13.36% overall, confirming that task specialization does not carry over to robust open-ended generation. The category pattern mirrors the MCQ results: top models do relatively better on orientation and some relative-position items (GPT-5-mini: Rel. Pos. 49.53%, Orientation 37.13%), while depth & occlusion, size & scale, and especially spatial navigation stay consistently hard, with many models below 30%. The open-ended setting therefore reinforces, in even starker numbers, that multi-step grounding and perceptual reasoning are the binding bottlenecks.

5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

The move from multiple-choice to open-ended answering carries a large and stable cost. Across the 25 models, the average MCQ→open-ended gap is 23.0% ($\sigma = 5.5\%$), and it is remarkably even across categories—subtask means fall in a tight 22.89% (spatial navigation) to 24.57% (3D geometry) band, so no category is exempt. The size of the drop, however, depends systematically on model type: specialist spatial reasoning models suffer the largest gaps (about 27% on average, reaching 36.68% in spatial navigation and 34.44% in orientation), whereas reasoning-oriented models shrink the gap to roughly 19% with lower subtask variance—direct evidence that instruction-tuning and CoT decoding stabilize generative outputs. Correlation analysis pins down where the disparity originates: spatial navigation is almost perfectly predictive of the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) close behind. A handful of negative or near-zero gaps (Llama-3.2-11B: −1.98 in depth & occlusion; o4-mini: −3.18 in relative position) run the other way, revealing that poorly designed MCQ distractors can understate—not just overstate—true competence. We attribute this format sensitivity to MCQ’s structural advantage, specialization bias toward categorical tasks, the sequential demands of spatio-navigation, and calibration differences in generative output. The practical upshot is quantitative and clear: reporting MCQ alone overstates real spatial competence by roughly 23 points on average. We therefore recommend pairing MCQ with open-ended evaluation, auditing distractors, employing stepwise generation or instruction-tuning, and reporting per-subtask diagnostics to capture true model performance and remove format-dependent artifacts. More details are in Appendix G and Table 18.

5.3 ERROR ANALYSIS

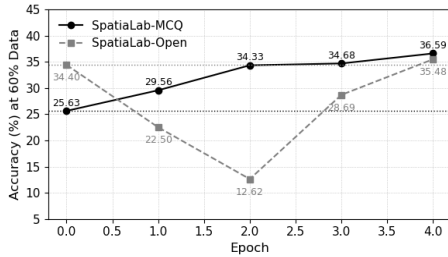
SPATIALAB-MCQ. Our subtask-level analysis (Appendix E) shows that high peaks coexist with steep collapses within the same model, so headline accuracy hides brittle competence. Closed-source models post the highest single-task scores (85.71% in *Stacking Orientation*) yet fall to near-chance on *Relative Size Comparison*—a swing of tens of points across two spatial subtasks. Open-source scaling lifts the ceiling (80.56% in *Corner/Angle Positioning*) but not the floor, with catastrophic failures as low as 2.0% in *Object Rotation*. Reasoning-augmented and spatially specialized models add only localized gains (e.g., in occlusion inference or navigation) and stay capped below 55% on physical-abstraction tasks such as *Gravity Effects* and *Stability Prediction*. Error distributions further reveal systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*), while stronger results in *Obstacle Avoidance* suggest shortcut exploitation rather than robust planning. Taken together, these patterns confirm that current VLMs rely heavily on surface-level correlations and lack stable encodings of orientation, physics, and compositional logic. Progress will likely require both richer training distributions (e.g., embodied and physics-driven data) and architectural mechanisms for geometric and reference-frame grounding.

SPATIALAB-OPEN. Error analysis of SPATIALAB-OPEN (Appendix F) exposes the same peak-and-collapse signature, only more extreme. The best closed-source model, GPT-5-mini, reaches 58.14% in *Directional Relations* yet collapses on *Proximity Gradients*, so its dominance is task-specific rather than uniform. Large open-source models hit 60.0% in *Depth & Occlusion* but plunge to 9.3% on *Proximity*—a roughly 50-point intra-model swing—confirming that scale alone does not buy robustness. Smaller models show isolated strengths (26.0% in *Relative Size Comparison*) alongside outright zeros (0.0% in *Betweenness*), the clearest illustration of compact-architecture limits. Reasoning-tuned systems are the most volatile: Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness* while peers on the same task fall to single digits, revealing fragile integration rather than a stable capability. The specialized spatial models remain worst overall, rarely exceeding 20% and often failing outright on core relational tasks. The failure distribution concentrates in occlusion handling, orientation, and multi-step relational chaining, whereas partial success on size-scale cues points to reliance on superficial correlations. Together these numbers implicate three root causes—missing representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines—and motivate unified geometric encodings, physics-aware reasoning, and embodied data to convert isolated peaks into consistent spatial competence.

Qualitative Error Analysis. To make the failure modes precise, we conducted a qualitative error analysis (see Appendix I) and found that errors are not random noise but concentrate into a small number of recurring classes. The dominant classes are spatial mislocalization (confusing referents in crowded scenes); perspective and scale mistakes, where object-size priors override image-based cues; and occlusion or ordering failures, especially with thin or partially hidden structures. We also see attribute confusion (mixing perceptual with functional properties) and open-ended rationalizations that read fluently but are visually ungrounded. Crucially, these classes recur across architectures and prompting styles, indicating structural biases in current VLMs rather than dataset-specific artifacts. The strongest quantitative pattern is that failure rates spike whenever multiple cues must be fused—for instance, depth ordering combined with relative size—and confidence calibration is worst in exactly the open-ended setting where accuracy is lowest. Our diagnostic protocols confirm that models routinely ignore minimal but decisive visual features, lean on brittle heuristics, and fail to update under counterfactual edits. These converging observations point to insufficient object-centric binding, missing geometric supervision, and training objectives that reward plausibility over grounding, and they show that current VLMs pair strong coarse perception with weak multi-cue integration—underscoring the need for geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Turning on a model’s built-in reasoning consistently helps: reasoning-enabled models beat their own non-reasoning baselines in both formats, with the largest single-category gain of +13.1% in Relative Positioning under MCQ. The benefit is real but uneven—it stabilizes open-ended performance yet actually declines in Size & Scale—so reasoning modules



(a) SFT accuracy trends over training epochs.

Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(b) SFT effect by category (accuracy %, Before vs. After, with Gain) for MCQ and Open-ended formats. SFT lifts MCQ accuracy in all six categories (overall 25.63 → 36.59, +10.97; largest gains in Size & Scale +17.11 and Spatial Navigation +16.08), whereas open-ended gains are near zero overall (+1.07) and even negative in Relative Positioning (-0.79) and Size & Scale (-3.95)—the +11-vs-+1 split that signals teaching-to-the-test. Positive gains are shaded green, regressions red.

Figure 4: SFT training results: (a) accuracy trends over epochs, and (b) final per-category accuracy, showing consistent MCQ gains but flat-to-negative open-ended transfer.

improve logical consistency without resolving grounding or scale sensitivity. Detailed per-category comparisons appear in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Prompted CoT is the clearest negative result: across models it delivers little benefit and usually lowers accuracy (Table 19), with overall changes such as -2.67 points for InternVL-3-78b (MCQ) and -18.00 points for Gemini-2.5-Flash (MCQ), and orientation the lone category showing consistent gains. Step-by-step reasoning thus helps directional alignment but backfires on tasks that need perceptual grounding—depth, scale, and spatial navigation—where it amplifies flawed priors rather than correcting them. The takeaway is quantitative: solving SPATIALAB requires perceptual understanding that logical chaining alone cannot supply (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Adding self-reflection to CoT—explicitly prompting the model to review and correct its own steps (see Figure 8 in the Appendix)—recovers some of that loss in the multiple-choice setting, most visibly for Gemini-2.5-Flash, whose overall MCQ accuracy rises by +20.66 points over plain CoT, with the strongest gains in geometry and depth. The effect does not carry to open-ended tasks (Appendix H.3), confirming that reflection cannot substitute for missing perceptual grounding and reinforcing the format-dependence seen throughout.

Supervised Fine-Tuning (SFT). SFT gives the largest and most systematic within-format gains we observe. Fine-tuning Qwen2.5-VL-3B-Instruct on a stratified 40% split of SPATIALAB (~560 samples) raises MCQ accuracy in every category (Figure 4), lifting the overall MCQ score from 25.63% to 36.59% (+10.97 points), with the biggest jumps in Size & Scale (+17.11) and Spatial Navigation (+16.08). The same model, however, barely moves on open-ended tasks (overall +1.07 points) and even regresses in Relative Positioning (-0.79) and Size & Scale (-3.95). This +11-vs-+1 split is direct evidence of overfitting to task-specific answer distributions—SFT that *teaches to the test* rather than fostering generalizable reasoning (Wang et al., 2025). The near-flat open-ended transfer points to representations tuned for categorical discrimination but unstable in generation, plausibly reflecting catastrophic forgetting of linguistic priors (Huang et al., 2024a) compounded by discrete-choice-aligned objectives. In short, SFT reliably boosts constrained-format accuracy while stalling progress on naturalistic spatial reasoning. More details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). Finally, we test SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) for structured spatial reasoning on images. It decomposes visual reasoning into specialized sub-tasks—object segmentation, attribute extraction, spatial relation mapping, and transformation tracking—and consolidates them into a unified structured representation, propagating low-level cues into higher-order relational inferences; we run it on Gemini-2.5-Flash at low temperature for deterministic multi-stage refinement. The results are the most polarized of any intervention: Orientation improves by +8.00% (MCQ) and a striking +36.00% (open-ended), yet Depth & Occlusion and Spatial Navigation drop by -24.00% and -12.00% in open evaluation. In other words, agentic decomposition delivers its single largest gain and two of its largest losses on the same benchmark, depending on the category. Where perceptual priors (depth, occlusion) are weak, multi-step reasoning entrenches misconceptions instead of fixing them. The pattern is consistent with every other intervention here: agentic workflows help tasks

reducible to sequential alignment but cannot manufacture the perceptual grounding that complex 3D scenes demand. More details are available in Appendix H.5.

6 DISCUSSION

Spatial reasoning is central to embodied intelligence, yet most benchmarks rely on synthetic scenes, templated questions, or narrow task scopes that overlook real-world visual variability. We introduce SPATIALAB, a benchmark of 1,400 visual question–answer pairs covering six major spatial categories and 30 subcategories, including 3D geometry, occlusion, and navigation. The benchmark uses both Multiple-Choice and Open-Ended formats to separate option-based pattern use from genuine generative reasoning, and reporting both on identical items lets us measure the format effect directly—an average 23-point MCQ→open-ended drop. Its quality is supported by a multi-stage review process and complexity checks covering lighting, texture, and spatial diversity. Evaluated under one protocol against a shared human baseline, more than 25 contemporary models cluster far below humans—no better than 54.93% in MCQ and 40.93% in open-ended against 87.57% and 64.93% for humans—so SPATIALAB functions as a precise, quantitative diagnostic for the weaknesses of current vision–language systems.

Detailed error analysis localizes systematic failures in transparency, multi-step navigation, and occlusion, indicating that models lean on surface-level texture cues rather than internalized 3D geometric representations. The intervention study sharpens this picture with numbers that point the same way. SFT delivered clear structured-format gains (+10.97 points overall in MCQ) but essentially no open-ended transfer (+1.07), a split consistent with “catastrophic forgetting” of linguistic priors when optimizing for discrete answers. Chain-of-Thought helped the sequential Orientation category yet on balance lowered accuracy (e.g., −18.00 points overall for Gemini-2.5-Flash in MCQ), amplifying rather than correcting perceptual errors in Depth and Scale. Agentic decomposition was the most polarized of all, adding +36.00% in open-ended Orientation while subtracting −24.00% in Depth & Occlusion. Read together, these consistent, sign-flipping results show that scaling and reasoning scaffolds alone cannot close the 24–33-point human gap; doing so will require architectural innovations that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

Spatial reasoning remains a core bottleneck for VLMs, particularly in visually complex, real-world settings. SPATIALAB introduces 1,400 diverse QA pairs across six spatial categories and thirty subcategories, enabling fine-grained, directly comparable evaluation via both MCQ and open-ended formats against a shared human baseline. Across more than 25 models and every performance-improving method we tried, the gap to humans never closes: the best model reaches 54.93% in MCQ and 40.93% in open-ended, versus 87.57% and 64.93% for humans—shortfalls of 32.64 and 24.00 points—and open-ended generation costs a further 10–25 points relative to multiple-choice. Error analysis pins the residual failures on depth, occlusion, 3D geometry, and multi-step navigation, reflecting reliance on superficial cues over grounded spatial representations. By quantifying these gaps on a common scale, SPATIALAB turns diffuse claims of “weak spatial reasoning” into measurable, per-category targets for progress. Our benchmark thus enables holistic evaluation and provides actionable, numeric insights for developing VLMs with human-aligned spatial understanding. Future progress will require richer training signals (e.g., embodied interaction, action-conditioned feedback) and architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We have taken multiple steps to ensure ethical compliance throughout the development of this work. All metadata associated with the images was removed to protect privacy, and any personally identifiable features (such as human faces or license plates) were minimized or excluded wherever possible. The dataset was carefully curated to avoid copyright concerns, with diligent attention paid to sourcing and licensing, though minor oversights may occur despite best efforts. The benchmark construction process was designed with transparency and fairness in mind, ensuring that no demographic groups

were singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To facilitate reproducibility, we provide a detailed description of the dataset construction process, annotation pipeline, and evaluation protocol in the main text and appendix. All filtering, quality-control, and validation steps are explicitly documented, with clear definitions of task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in the appendix to support consistent replication of our results.

REFERENCES

- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022.
- Anthropic. Model Card Addendum: Claude 3.5 Haiku and Upgraded Claude 3.5 Sonnet, October 2024. URL <https://assets.anthropic.com/m/1cd9d098ac3e6467/original/Claude-3-Model-Card-October-Addendum.pdf>. Accessed: 2025-09-11.
- Daichi Azuma, Taiki Miyanishi, Shuhei Kurita, and Motoaki Kawanabe. Scanqa: 3d question answering for spatial scene understanding. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 19129–19139, 2022.
- Alan Baddeley. Working memory. *Comptes Rendus de l’Académie des Sciences-Series III-Sciences de la Vie*, 321(2-3):167–173, 1998.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. URL <https://arxiv.org/abs/2502.13923>.
- Yoav Bar-Anan, Nira Liberman, and Yaacov Trope. The association between psychological distance and construal level: evidence from an implicit association test. *Journal of experimental psychology: General*, 135(4):609, 2006.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil J. Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong T. Tran, Vincent Vanhoucke, Steve Vega, Quan Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: robotics transformer for real-world control at scale. In *Robotics: Science and Systems XIX, Daegu, Republic of Korea, July 10-14, 2023*, 2023.
- Wenxiao Cai, Yaroslav Ponomarenko, Jianhao Yuan, Xiaoqi Li, Wankou Yang, Hao Dong, and Bo Zhao. Spatialbot: Precise spatial understanding with vision language models. *CoRR*, abs/2406.13642, 2024.
- Zhongang Cai, Yubo Wang, Qingping Sun, Ruishi Wang, Chenyang Gu, Wanqi Yin, Zhiqian Lin, Zhitao Yang, Chen Wei, Xuanke Shi, Kewang Deng, Xiaoyang Han, Zukai Chen, Jiaqi Li, Xiangyu Fan, Hanming Deng, Lewei Lu, Bo Li, Ziwei Liu, Quan Wang, Dahua Lin, and Lei Yang. Has gpt-5 achieved spatial intelligence? an empirical study, 2025. URL <https://arxiv.org/abs/2508.13142>.

- Christopher F Chabris, Thomas E Jerde, Anita W Woolley, Margaret E Gerbasi, Jonathon P Schuldt, Sean L Bennett, J Richard Hackman, and Stephen M Kosslyn. Spatial and object visualization cognitive styles: Validation studies in 3800 individuals. *Group brain technical report*, 2(1-20):2, 2006.
- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh, Leonidas Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. *arXiv preprint arXiv:2401.12168*, 2024a. URL <https://arxiv.org/abs/2401.12168>.
- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Dorsa Sadigh, Leonidas J. Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pp. 14455–14465. IEEE, 2024b.
- Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. *CoRR*, abs/2503.01773, 2025.
- An-Chieh Cheng, Hongxu Yin, Yang Fu, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, and Sifei Liu. Spatialrgpt: Grounded spatial reasoning in vision-language models. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024.
- Open X.-Embodiment Collaboration, Abhishek Padalkar, Acorn Pooley, Ajinkya Jain, Alex Bewley, Alexander Herzog, Alex Irpan, Alexander Khazatsky, Anant Raj, Anikait Singh, Anthony Brohan, Antonin Raffin, Ayzaan Wahid, Ben Burgess-Limerick, Beomjoon Kim, Bernhard Schölkopf, Brian Ichter, Cewu Lu, Charles Xu, Chelsea Finn, Chenfeng Xu, Cheng Chi, Chenguang Huang, Christine Chan, Chuer Pan, Chuyuan Fu, Coline Devin, Danny Driess, Deepak Pathak, Dhruv Shah, Dieter Buehler, Dmitry Kalashnikov, Dorsa Sadigh, Edward Johns, Federico Ceola, Fei Xia, Freek Stulp, Gaoyue Zhou, Gaurav S. Sukhatme, Gautam Salhotra, Ge Yan, Giulio Schiavi, Gregory Kahn, Hao Su, Haoshu Fang, Haochen Shi, Heni Ben Amor, Henrik I. Christensen, Hiroki Furuta, Homer Walke, Hongjie Fang, Igor Mordatch, Ilija Radosavovic, and et al. Open x-embodiment: Robotic learning datasets and RT-X models. *CoRR*, abs/2310.08864, 2023.
- Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, September 1951. ISSN 1860-0980. doi: 10.1007/bf02310555. URL <http://dx.doi.org/10.1007/BF02310555>.
- Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas A. Funkhouser, and Matthias Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*, pp. 2432–2443. IEEE Computer Society, 2017.
- Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. Palm-e: An embodied multimodal language model. In *ICML*, 2023.
- Chris Drummond. Replicability is not reproducibility: nor is it good science. In *Proceedings of the Evaluation Methods for Machine Learning Workshop at the 26th ICML*, volume 1, Montreal, Canada, 2009. National Research Council of Canada.
- Chris Drummond and Nathalie Japkowicz. Warning: statistical benchmarking is addictive. kicking the habit in machine learning. *Journal of Experimental and Theoretical Artificial Intelligence*, 22(1):67–80, March 2010. ISSN 1362-3079. doi: 10.1080/09528130903010295. URL <http://dx.doi.org/10.1080/09528130903010295>.

- Mengfei Du, Binhao Wu, Zejun Li, Xuanjing Huang, and Zhongyu Wei. Embspatial-bench: Benchmarking spatial understanding for embodied tasks with large vision-language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024 - Short Papers, Bangkok, Thailand, August 11-16, 2024*, pp. 346–355. Association for Computational Linguistics, 2024.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: multimodal large language models can see but not perceive. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol (eds.), *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XXIII*, volume 15081 of *Lecture Notes in Computer Science*, pp. 148–166. Springer, 2024.
- Howard E Gardner. *Frames of mind: The theory of multiple intelligences*. Basic books, 2011.
- Gemma Team, Google DeepMind. Gemma 3 technical report, 2025. URL <https://storage.googleapis.com/deepmind-media/gemma/Gemma3Report.pdf>. Accessed: 2025-09-11.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelier van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shao-liang Nie, Sharan Narang, Sharath Rapparthi, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie DelPierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei

756 Baeviski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu,
 757 Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit
 758 Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury,
 759 Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer,
 760 Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu,
 761 Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido,
 762 Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu
 763 Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer,
 764 Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu,
 765 Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc
 766 Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily
 767 Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers,
 768 Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank
 769 Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Sweet,
 770 Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan,
 771 Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph,
 772 Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog,
 773 Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James
 774 Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny
 775 Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings,
 776 Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai
 777 Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik
 778 Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle
 779 Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng
 780 Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish
 781 Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim
 782 Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle
 783 Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang,
 784 Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam,
 785 Natasha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier,
 786 Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia
 787 Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro
 788 Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani,
 789 Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy,
 790 Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin
 791 Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Sibi, Sai Jayesh Bondu,
 792 Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh
 793 Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay,
 794 Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang,
 795 Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie
 796 Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta,
 797 Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman,
 798 Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun
 799 Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria
 800 Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru,
 801 Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz,
 802 Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv
 803 Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi,
 804 Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait,
 805 Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The
 llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.
 806 Tanmay Gupta and Aniruddha Kembhavi. Visual programming: Compositional visual reasoning
 807 without training. In *CVPR*, 2023.
 808 Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David
 809 Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti
 Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández
 del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy,
 Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming

with NumPy. *Nature*, 585(7825):357–362, September 2020. doi: 10.1038/s41586-020-2649-2. URL <https://doi.org/10.1038/s41586-020-2649-2>.

Md Tanzib Hosain, Salman Rahman, Md Kishor Morol, and Md Rizwan Parvez. Xolver: Multi-agent reasoning with holistic experience learning just like an olympiad team, 2025. URL <https://arxiv.org/abs/2506.14234>.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.

Jianheng Huang, Leyang Cui, Ante Wang, Chengyi Yang, Xinting Liao, Linfeng Song, Junfeng Yao, and Jinsong Su. Mitigating catastrophic forgetting in large language models with self-synthesized rehearsal. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1416–1428, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.77. URL <https://aclanthology.org/2024.acl-long.77/>.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *CoRL*, 2022.

Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *CoRL*, 2024b.

Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pp. 6700–6709. Computer Vision Foundation / IEEE, 2019.

Mengdi Jia, Zekun Qi, Shaochen Zhang, Wenyao Zhang, Xinqiang Yu, Jiawei He, He Wang, and Li Yi. Omnispatial: Towards comprehensive spatial reasoning benchmark for vision language models, 2025. URL <https://arxiv.org/abs/2506.03135>.

Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning, 2016. URL <https://arxiv.org/abs/1612.06890>.

Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s ”up” with vision-language models? investigating their struggle with spatial reasoning. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023*, pp. 9161–9175. Association for Computational Linguistics, 2023.

Terry K. Koo and Mae Y. Li. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2):155–163, June 2016. ISSN 1556-3707. doi: 10.1016/j.jcm.2016.02.012. URL <http://dx.doi.org/10.1016/j.jcm.2016.02.012>.

Eliza Kosoy, Annya Dahmani, Andrew K. Lampinen, Iulia M. Comsa, Soojin Jeong, Ishita Dasgupta, and Kelsey Allen. Decoupling the components of geometric understanding in vision language models. *CoRR*, abs/2503.03840, 2025.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. BLIP-2: bootstrapping language-image pre-training with frozen image encoders and large language models. *CoRR*, abs/2301.12597, 2023.

Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. *Trans. Assoc. Comput. Linguistics*, 11:635–651, 2023.

Siqi Liu, Ian Gemp, Luke Marris, Georgios Piliouras, Nicolas Heess, and Marc Lanctot. Re-evaluating open-ended evaluation of large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=kbOAIXKWgx>.

-
- Yao Mu, Qinglong Zhang, Mengkang Hu, Wenhai Wang, Mingyu Ding, Jun Jin, Bin Wang, Jifeng Dai, Yu Qiao, and Ping Luo. Embodiedgpt: Vision-language pre-training via embodied chain of thought. In *NeurIPS*, 2023.
- OpenAI. Gpt-4v(ision) system card, 2023. URL <https://openai.com/research/gpt-4v-system-card>.
- OpenAI. Gpt-4o system card, 2024. URL <https://arxiv.org/abs/2410.21276>.
- OpenAI. Openai o3 and o4-mini system card, April 2025. URL <https://openai.com/index/o3-o4-mini-system-card/>. System Cards. Accessed: 2025-09-11.
- Stephen Palmer. *Vision Science: From Photons to Phenomenology*, volume 1. MIT Press, 01 1999.
- Atin Pothiraj, Elias Stengel-Eskin, Jaemin Cho, and Mohit Bansal. Capture: Evaluating spatial reasoning in vision language models via occluded object counting. *CoRR*, abs/2504.15485, 2025.
- Fred H Previc. The neuropsychology of 3-d space. *Psychological bulletin*, 124(2):123, 1998.
- Zekun Qi, Runpei Dong, Shaochen Zhang, Haoran Geng, Chunrui Han, Zheng Ge, Li Yi, and Kaisheng Ma. Shapellm: Universal 3d object understanding for embodied interaction. In *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XLIII*, volume 15101 of *Lecture Notes in Computer Science*, pp. 214–238. Springer, 2024.
- Zekun Qi, Wenyao Zhang, Yufei Ding, Runpei Dong, Xinqiang Yu, Jingwen Li, Lingyun Xu, Baoyu Li, Xialin He, Guofan Fan, Jiazhao Zhang, Jiawei He, Jiayuan Gu, Xin Jin, Kaisheng Ma, Zhizheng Zhang, He Wang, and Li Yi. Sofar: Language-grounded orientation bridges spatial reasoning and object manipulation. *CoRR*, abs/2502.13143, 2025.
- Tiago Ramalho, Tomás Kociský, Frederic Besse, S. M. Ali Eslami, Gábor Melis, Fabio Viola, Phil Blunsom, and Karl Moritz Hermann. Encoding spatial relations from natural language. *CoRR*, abs/1807.01670, 2018.
- Fatemeh Shiri, Xiao-Yu Guo, Mona Far, Xin Yu, Reza Haf, and Yuan-Fang Li. An empirical analysis on spatial reasoning capabilities of large multimodal models. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024, Miami, FL, USA, November 12-16, 2024*, pp. 21440–21455. Association for Computational Linguistics, 2024.
- Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428, 1979. ISSN 0033-2909. doi: 10.1037/0033-2909.86.2.420. URL <http://dx.doi.org/10.1037/0033-2909.86.2.420>.
- Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield. Robospatial: Teaching spatial understanding to 2d and 3d vision-language models for robotics. *CoRR*, abs/2411.16537, 2024.
- StepFun. Step-3 is large yet affordable: Model-system co-design for cost-effective decoding, 2025. URL <https://arxiv.org/abs/2507.19427>.
- Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsafaris. Mind the gap: Benchmarking spatial reasoning in vision-language models. *CoRR*, abs/2503.19707, 2025.
- Emilia Szymanska, Mihai Dusmanu, Jan-Willem Burchfiel, Mahdi Rad, and Marc Pollefeys. Space3d-bench: Spatial 3d question answering benchmark. *CoRR*, abs/2408.16662, 2024.
- Mohsen Tavakol and Reg Dennick. Making sense of cronbach’s alpha. *International Journal of Medical Education*, 2:53–55, June 2011. ISSN 2042-6372. doi: 10.5116/ijme.4dfb.8dfd. URL <http://dx.doi.org/10.5116/ijme.4dfb.8dfd>.
- Gemini Team. Gemini: A family of highly capable multimodal models, 2025a. URL <https://arxiv.org/abs/2312.11805>.

- Kimi Team. Kimi-vl technical report, 2025b. URL <https://arxiv.org/abs/2504.07491>.
- V Team, Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, Shuaiqi Duan, Weihang Wang, Yan Wang, Yean Cheng, Zehai He, Zhe Su, Zhen Yang, Ziyang Pan, Aohan Zeng, Baoxu Wang, Bin Chen, Boyan Shi, Changyu Pang, Chenhui Zhang, Da Yin, Fan Yang, Guoqing Chen, Jiazheng Xu, Jiale Zhu, Jiali Chen, Jing Chen, Jinhao Chen, Jinghao Lin, Jinjiang Wang, Junjie Chen, Leqi Lei, Letian Gong, Leyi Pan, Mingdao Liu, Mingde Xu, Mingzhi Zhang, Qinkai Zheng, Sheng Yang, Shi Zhong, Shiyu Huang, Shuyuan Zhao, Siyan Xue, Shangqin Tu, Shengbiao Meng, Tianshu Zhang, Tianwei Luo, Tianxiang Hao, Tianyu Tong, Wenkai Li, Wei Jia, Xiao Liu, Xiaohan Zhang, Xin Lyu, Xinyue Fan, Xuancheng Huang, Yanling Wang, Yadong Xue, Yanfeng Wang, Yanzi Wang, Yifan An, Yifan Du, Yiming Shi, Yiheng Huang, Yilin Niu, Yuan Wang, Yuanchang Yue, Yuchen Li, Yutao Zhang, Yuting Wang, Yu Wang, Yuxuan Zhang, Zhao Xue, Zhenyu Hou, Zhengxiao Du, Zihan Wang, Peng Zhang, Debing Liu, Bin Xu, Juanzi Li, Minlie Huang, Yuxiao Dong, and Jie Tang. Glm-4.5v and glm-4.1v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning, 2025. URL <https://arxiv.org/abs/2507.01006>.
- Yaacov Trope and Nira Liberman. Construal-level theory of psychological distance. *Psychological review*, 117(2):440, 2010.
- Raphael Vallat. Pingouin: statistics in python. *Journal of Open Source Software*, 3(31):1026, November 2018. doi: 10.21105/joss.01026.
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Yubo Wang, Xiang Yue, and Wenhui Chen. Critique fine-tuning: Learning to critique is more effective than learning to imitate, 2025. URL <https://arxiv.org/abs/2501.17703>.
- Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in space: How multimodal large language models see, remember, and recall spaces. *CoRR*, abs/2412.14171, 2024.
- Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, and Lijuan Wang. MM-REACT: prompting chatgpt for multimodal reasoning and action. *CoRR*, abs/2303.11381, 2023.
- Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, and Dieter Fox. Robopoint: A vision-language model for spatial affordance prediction for robotics. *CoRR*, abs/2406.10721, 2024.
- Kaizhi Zheng, Xiaotong Chen, Odest Chadwicke Jenkins, and Xin Eric Wang. Vlmbench: A compositional benchmark for vision-and-language manipulation, 2022. URL <https://arxiv.org/abs/2206.08522>.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Nianchen Deng, Songze Li, Yinan He, Tan Jiang, Jiapeng Luo, Yi Wang, Conghui He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Yingdong Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Han Lv, Lijun Wu, Kaipeng Zhang, Huipeng Deng, Jiaye Ge, Kai Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhui Wang. InternV3: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. URL <https://arxiv.org/abs/2504.10479>.

972 Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart,
973 Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong T. Tran, Radu Soricut,
974 Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S.
975 Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu,
976 Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov,
977 Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol
978 Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava
979 Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar,
980 Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han.
981 RT-2: vision-language-action models transfer web knowledge to robotic control. In *Conference on*
982 *Robot Learning, CoRL 2023, 6-9 November 2023, Atlanta, GA, USA*, volume 229 of *Proceedings*
983 *of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

1026	A Related Work	22
1027	A.1 Detailed Analysis of Limitations in Prior Benchmarks	23
1028		
1029	B Task Design and Descriptions	24
1030	B.1 Relative Positioning	25
1031	B.2 Depth & Occlusion	25
1032	B.3 Orientation	25
1033	B.4 Size & Scale	26
1034	B.5 Spatial Navigation	26
1035	B.6 3D Geometry	26
1036		
1037	C More Details on Benchmark Construction	27
1038	C.1 Annotator Recruitment	27
1039	C.2 Annotator Training and Protocol	27
1040		
1041	D More details on Experiments and Evaluation	28
1042	D.1 Multiple Choice Evaluation	28
1043	D.2 Open-Ended Evaluation	28
1044	D.2.1 Evaluation of LLM-as-a-Judge for Open-ended Evaluation	28
1045		
1046	E Error Analysis of SPATIALAB-MCQ	29
1047	E.1 Sub-Category-wise Quantitative Error Analysis	29
1048	E.2 Model Failure Analysis	34
1049		
1050	F Error Analysis of SPATIALAB-OPEN	34
1051	F.1 Sub-Category-wise Quantitative Error Analysis	34
1052	F.2 Model Failure Analysis	38
1053		
1054	G Analysis of MCQ vs. Open-Ended Performance Gaps	39
1055	G.1 Quantitative snapshot	40
1056	G.2 Patterns and observations	40
1057	G.3 Hypotheses on root causes	41
1058	G.4 Practical implications	42
1059		
1060	H Details on Improving Visual Reasoning Capabilities	42
1061	H.1 Inherent Reasoning Mechanisms	42
1062	H.2 Chain-of-Thought (CoT) Prompting	42
1063	H.3 Chain-of-Thought (CoT) Prompting with Self-Reflection	43
1064	H.4 Supervised Fine-Tuning (SFT)	45
1065	H.4.1 Setup and Implementation	45
1066	H.4.2 Supervised Fine-Tuning (SFT) Performance	45
1067		
1068		
1069		
1070		
1071		
1072		
1073		
1074		
1075		
1076		
1077		
1078		
1079		

1080		
1081	H.4.3	SFT Dynamics Analysis 47
1082	H.4.4	Performance Gain on External Benchmarks 48
1083	H.5	AI Agents for Spatial Reasoning 48
1084	H.5.1	Agents 48
1085	H.5.2	Technical Implementation Details 50
1086	H.5.3	Evaluation and Analysis 50
1087		
1088		
1089	I	Qualitative Error Analysis and Error Patterns 51
1090	I.1	Overview 51
1091	I.2	Error taxonomy and analysis 51
1092	I.2.1	Spatial mislocalization and reference confusion. 51
1093	I.2.2	Perspective and scale mistakes. 52
1094	I.2.3	Occlusion and ordering failures. 52
1095	I.2.4	Attribute confusion and semantic swap. 52
1096	I.2.5	Open-ended rationalization without verification. 53
1097	I.3	Error patterns and cross-model signals 53
1098	I.4	Mitigation strategies 54
1099		
1100	J	Representative Benchmark Samples Across All Sub-Categories 58
1101		
1102	K	Statistical Robustness and Dataset Stability 88
1103	K.1	Model Robustness 88
1104	K.1.1	Multiple Run Averages and Deviations 88
1105	K.1.2	Intra-Class Correlation (ICC) 88
1106	K.2	Dataset Stability and Internal Consistency 88
1107	K.2.1	Resampling Study 89
1108	K.2.2	Item-Level Consistency 89
1109	K.2.3	Cronbach’s Alpha 89
1110	K.3	Empirical Results 90
1111	K.3.1	Model Evaluation Robustness 90
1112	K.3.2	Dataset Stability and Internal Consistency 91
1113	K.3.3	Overall Discussion 91
1114		
1115		
1116		
1117		
1118		
1119		
1120		
1121		
1122		
1123		
1124		
1125		
1126		
1127		
1128		
1129		
1130		
1131		
1132		
1133		

A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning, the capacity to understand geometric structures and relational layouts in visual scenes, is central to embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) tested compositional reasoning with synthetic or curated templates, which allowed controlled analysis but provided limited ecological validity. With the advent of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), spatial reasoning has been revisited in more naturalistic datasets, yet a consistent quantitative pattern persists: models that score highly on captioning and visual QA drop sharply on tasks requiring depth ordering, occlusion inference, or consistent reference-frame alignment. This recurring gap between coarse recognition and precise spatial inference is the empirical motivation for our study, and it underlines a broader point—scale alone does not guarantee progress in structured spatial understanding.

Spatial reasoning benchmarks. A range of benchmarks have recently been proposed to probe these gaps, and Table 1 lets us compare them on task coverage, scale, difficulty, and best-baseline accuracy side by side. SCANQA (Azuma et al., 2022) was among the first embodied QA datasets, requiring reasoning about 3D indoor environments, but its limited scale (800 scenes) and easy (E) complexity restrict broader generalization. Datasets such as VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) expanded scope to real-world or household contexts, yet relied on template-driven questions—reducing linguistic variability and overrepresenting simple binary or multiple-choice relations—and their best baselines already reach 70.10% and 60.40%, leaving little headroom. More recent efforts, EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), add task variety but stay small (2.2K and 211 samples) with template-heavy or narrow manual annotation. SPATIALRGPT-BENCH and BLINK-SPATIAL widen domains and reach up to 12–14 categories, improving diversity, but still emphasize static snapshots or limited multi-view simulation and report GPT-4o baselines of 57.83% and 59.03%.

Embodiment, difficulty, and diversity. The trend toward embodiment is visible in datasets like ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions but whose template-based annotation makes questions easier to game with heuristic shortcuts—reflected in a high 71.90% best baseline. SPATIALVLM (Chen et al., 2024b) offers even greater scale (billions of QA pairs) yet covers only two categories, making it unsuitable for fine-grained diagnosis. At the opposite extreme, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) improve annotation fidelity but include only 288 scenes, too few for robust cross-model comparison. The pattern across the table is that scale and difficulty trade off: most existing benchmarks cap at “medium” complexity (Mm in Table 1) and rarely reach the “hard” or “extremely hard” levels that demand multi-step causal inference. SPATIALAB is the only entry that combines a large item count (1.2K+) with high-to-very-high difficulty, closing exactly this evaluation gap for next-generation models.

Evaluation protocols. Equally important are evaluation type and analysis depth. Most prior work employs close-ended QA or binary judgments (e.g., VISUAL SPATIAL, EMBSPATIAL-BENCH), which simplify scoring but let models inflate accuracy through option elimination rather than genuine reasoning. A few datasets—ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN—add open-ended evaluation, but only SPATIALAB-OPEN pairs it with very-high complexity, manual puzzle-style annotation, and comprehensive error analysis. Reporting both formats on the same items is what makes the 23-point average MCQ→open-ended drop measurable, exposing systematic weaknesses—including calibration and cross-model divergence—that single-format benchmarks cannot see. Without this granularity, headline gains can mislead, since models may be overfitting to dataset-specific biases rather than acquiring robust spatial competence.

Positioning of SPATIALAB. SPATIALAB directly addresses these limitations, and the contrast is quantitative. Where prior datasets span 2–14 task categories, SPATIALAB covers 30 across geometry, occlusion, orientation, relative positioning, size and scale, and navigation—roughly double the breadth of the widest prior suite. All questions are manually authored and puzzle-inspired rather than template-generated, which raises linguistic richness and cuts shortcut opportunities. The benchmark is also fully embodied and multi-modal over static and dynamic contexts, bridging robotics-style datasets (ROBOSPATIAL) and web-scale corpora (SPATIALVLM). The clearest evidence of difficulty

is the best-baseline column of Table 1: while earlier benchmarks admit baselines of 45–72%, the strongest models on SPATIALAB—InternVL3.5-72B and GPT-5-mini—reach only 54.93% (MCQ) and 40.93% (open-ended), the lowest in the table. This substantially larger unsolved margin is precisely what makes the benchmark useful for driving future research.

Toward next-generation evaluation. By combining high difficulty, the broadest coverage (30 categories), manual annotation, and open-ended tasks, SPATIALAB shifts the landscape of spatial reasoning benchmarks from template-driven, near-saturated evaluations toward comprehensive, embodied testing with ample unsolved margin. This design makes it possible not only to score correctness but to quantify systematic error patterns, probe causal and counterfactual reasoning, and measure calibration reliability. It thereby provides a rigorous foundation for the next phase of multimodal research, complementing datasets like MIND THE GAP (best baseline 48.83%) and VLM4D (62.00%) that explore temporal or generative aspects but remain narrower in coverage. We expect future work to use SPATIALAB as a high-bar diagnostic, pushing models beyond surface-level recognition toward genuinely grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To situate SPATIALAB within the broader landscape, we conducted a detailed examination of existing spatial reasoning benchmarks. Earlier efforts contributed important foundations, yet they share three structural constraints that inflate their reported accuracies and limit their value for evaluating current VLMs: synthetic-domain bias, limited task scope, and restrictive evaluation formats. Table 4 organizes the major limitations along these axes and contrasts each against SPATIALAB.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rely on synthetic environments or template-generated linguistic structures. Although these settings offer controlled conditions for isolating compositional reasoning, the visual domain lacks real-world complexity and the linguistic space is narrow. Models can rely on template regularities rather than grounded spatial interpretation. Related simulated benchmarks such as ROBOSPATIAL face similar issues, resulting in a large sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but remain restricted to indoor scenes. This prevents assessment in outdoor or mixed environments that are essential for embodied agents. High-quality manual datasets such as VSI-Bench offer richer annotation but remain small in scale, limiting their usefulness for high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack detailed taxonomies, making them unsuitable for diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, rely heavily on Binary or MCQ formats. These formats can inflate performance by enabling option elimination rather than requiring generative spatial reasoning. SPATIALAB mitigates these issues by combining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations of representative prior spatial benchmarks compared with SPATIALAB, organized by domain constraints, evaluation format, and task taxonomy. SPATIALAB is the only row that is simultaneously in-the-wild, dual-format (MCQ + Open), and fine-grained (30 subcategories), directly addressing the constraints that inflate accuracy on the other rows.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale (< 1K scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Understanding spatial reasoning requires more than simple recognition of objects; it depends on how those objects relate, align, and interact in physical space. To capture this complexity, we introduce *SpatialLab*, a benchmark designed to test fine-grained aspects of spatial understanding across diverse categories. The tasks in SpatialLab reflect how humans naturally describe and reason about environments, ranging from relative positioning (e.g., *What's to the left of the microwave?*) to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category is further broken into subcategories that highlight subtle but essential dimensions of reasoning, such as occlusion, orientation, navigation, and physical effects like gravity. By framing each subtask in accessible, everyday terms, SpatialLab encourages models to go beyond object detection and engage in structured spatial interpretation. This design allows us to probe how well systems generalize across contexts that matter for robotics, AR/VR, autonomous driving, and human-computer interaction. The following sections provide short, task-level descriptions for every category and subcategory in SpatialLab.

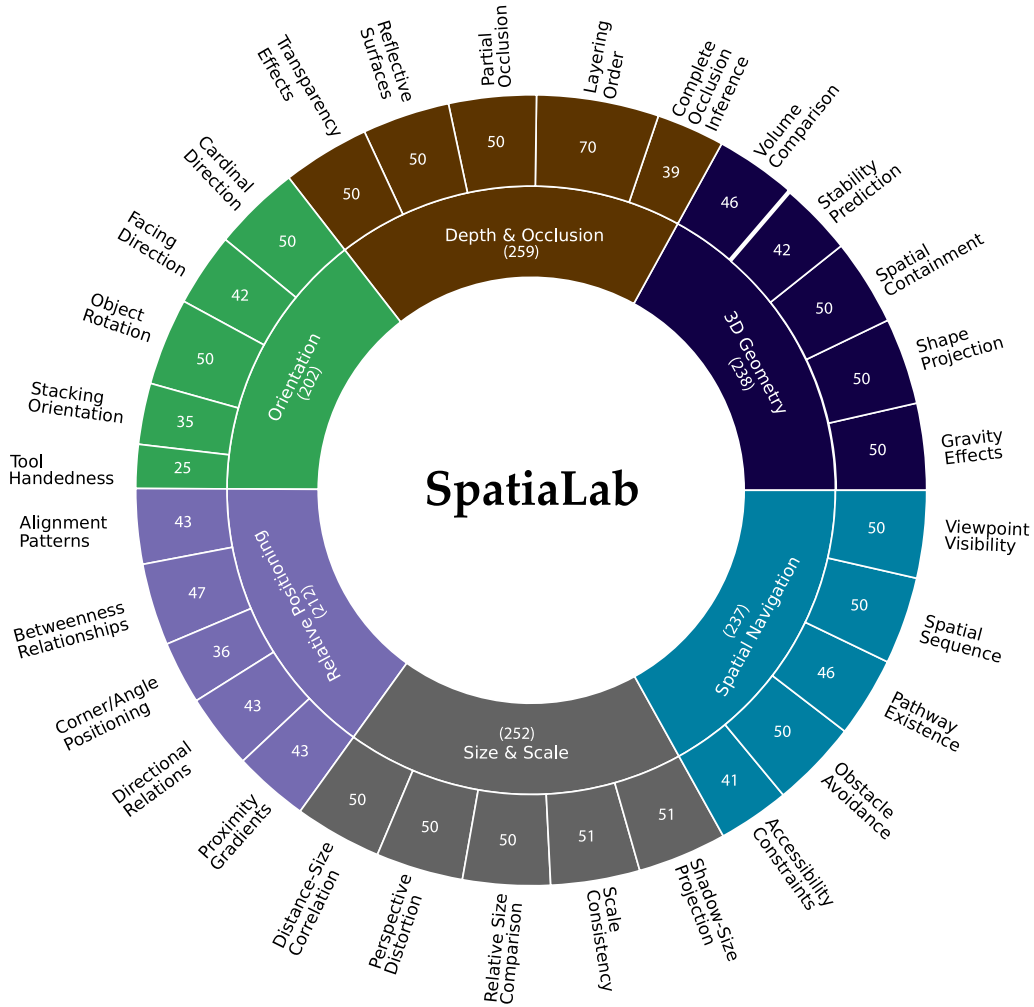


Figure 5: **Categories and subcategories of spatial reasoning in SPATIALAB.** Each category decomposes into five subcategories, yielding thirty task types in total.

The following outlines the rationale and real-world utility guiding each task design, per category:

1296 B.1 RELATIVE POSITIONING

1297
1298 **Directional Relations.** Directional relations describe how objects are positioned relative to one
1299 another, such as left, right, above, or below. Understanding these relations enables models to interpret
1300 spatial layouts more naturally, just like how humans describe scenes. For example, identifying what
1301 lies immediately to the left of an appliance helps in scene understanding and robot navigation.

1302 **Proximity Gradients.** Proximity reasoning involves recognizing which objects are closest or farthest
1303 from a reference point. This is key for questions like “Which object is nearest to the doorway?”
1304 Models capable of estimating gradients of closeness can support navigation tasks and assistive
1305 technologies.

1306 **Alignment Patterns.** Alignment concerns whether objects follow a certain orientation or order, such
1307 as being stacked vertically or arranged in a straight line. Recognizing alignment patterns helps in
1308 understanding organization in environments like shelves, tables, or warehouses. Such reasoning is
1309 also critical for tasks like robotic sorting and automated inspection.

1310 **Betweenness Relationships.** Betweenness focuses on identifying objects that occupy the middle
1311 ground between two references. For instance, asking what sits between the lamp and the armchair
1312 requires reasoning about spatial continuity. This helps in navigation, retrieval, and situational
1313 awareness tasks.

1314 **Corner/Angle Positioning.** Corners and angled positions often serve as reference anchors in spatial
1315 reasoning. Questions such as “What’s positioned in the back-right corner?” require the model to
1316 localize objects within constrained regions. This kind of reasoning is vital in robotics for mapping
1317 and placement.

1318 1319 1320 B.2 DEPTH & OCCLUSION

1321
1322 **Layering Order.** Layering requires distinguishing which objects appear in front versus behind. For
1323 example, identifying the frontmost item on a table reflects depth reasoning. This is particularly useful
1324 in AR/VR applications where visual stacking affects user experience.

1325 **Partial Occlusion.** Partial occlusion occurs when one object hides part of another from view.
1326 Recognizing the hidden item behind a coffee cup, for instance, requires filling in missing information.
1327 This strengthens robustness in perception systems operating in cluttered environments.

1328 **Complete Occlusion Inference.** Sometimes objects are fully hidden, yet clues like shadows or
1329 bulges reveal their presence. Inferring what lies under a blanket requires the model to use indirect
1330 evidence. Such reasoning mirrors human inference in uncertain visual conditions.

1331 **Transparency Effects.** Transparent surfaces, such as glass or water, allow partial visibility of objects
1332 behind them. Understanding what’s visible through a fish tank requires blending occlusion reasoning
1333 with transparency perception. This plays an important role in simulation, graphics, and embodied AI.

1334 **Reflective Surfaces.** Reflective reasoning asks models to interpret mirrors, shiny metals, or glass
1335 reflections. Identifying what is reflected in a mirror involves indirect spatial mapping. This is essential
1336 for safety in autonomous vehicles and situational awareness in robotics.

1337 1338 1339 B.3 ORIENTATION

1340
1341 **Cardinal Direction.** Cardinal reasoning relates to compass directions like north, south, east, and
1342 west. Asking which way a front door faces requires integrating global orientation cues. This is critical
1343 for navigation and geospatial alignment.

1344 **Object Rotation.** Rotation reasoning measures how much an object has been turned around an axis.
1345 Estimating that a chair is rotated by some degrees is crucial for accurate 3D modeling. This helps
1346 robots align objects correctly for manipulation.

1347 **Facing Direction.** Facing direction specifies the orientation of an object’s front side. For example, de-
1348 termining whether a monitor faces north requires mapping its surface orientation to global references.
1349 This is important in surveillance and robotics.

Stacking Orientation. Stacking involves understanding whether objects are piled vertically, horizontally, or in mixed orientations. Recognizing such layouts ensures stability in warehouses and homes. It is also critical for safe robotic stacking.

Tool Handedness. Certain tools, like scissors or gloves, are designed for left- or right-handed use. Identifying handedness requires fine-grained shape and orientation perception. This enhances assistive robotics in human-centered environments.

B.4 SIZE & SCALE

Relative Size Comparison. Relative comparisons assess whether one object is larger, taller, or smaller than another. Asking if a vase is taller than a statue requires proportional reasoning. This is crucial for packing, fitting, and recognition tasks.

Perspective Distortion. Perspective can make distant objects appear smaller than nearby ones, even if the actual size differs. Identifying illusions like a faraway car looking smaller than a nearby bicycle requires compensating for depth cues. This ability makes models more robust to camera perspectives.

Distance-Size Correlation. Here, size judgments depend on distance estimation. For instance, a nearby bicycle appearing larger than a far car should not confuse true size reasoning. This helps in navigation, AR/VR, and visual grounding.

Scale Consistency. Scale reasoning involves checking whether object sizes are reasonable in context. A mouse larger than a chair signals inconsistency, which models must detect. This strengthens realism in simulated environments.

Shadow-Size Projection. Shadows provide indirect cues about object size and light source position. Estimating how tall a light must be based on shadow length requires geometric reasoning. This is useful in graphics, architecture, and physics-based modeling.

B.5 SPATIAL NAVIGATION

Pathway Existence. Pathway reasoning checks if a clear route exists between two points. For example, determining if one can reach the desk without moving chairs tests navigability. This ability underpins indoor robot planning.

Obstacle Avoidance. Avoidance requires finding routes that minimize collisions. Asking for the clearest path from door to window directly maps to robotic and human navigation. It is essential in autonomous driving and indoor mobility aids.

Viewpoint Visibility. From a given viewpoint, certain objects may be blocked or visible. For example, checking what is obscured when sitting on the couch requires simulating perspective. This strengthens AR/VR realism and robotic vision.

Spatial Sequence. Sequence reasoning identifies the order of encounters along a path. Walking from the kitchen to the balcony, one might pass the dining table first. Such reasoning mimics human navigation memory.

Accessibility Constraints. Constraints identify which objects are blocked from access due to obstacles. For instance, a cabinet may be unreachable because of a table in front. This is crucial for assistive AI and ergonomic design.

B.6 3D GEOMETRY

Volume Comparison. Comparing volume asks which object occupies more three-dimensional space. Judging whether a box is larger than a sphere helps in packing and containment tasks. This is important for logistics and simulation.

Stability Prediction. Stability reasoning predicts whether adding or moving objects will topple a stack. For instance, checking if placing a book will collapse a pile mirrors human intuition. This supports safety in robotics and construction.

Shape Projection. Projection tasks ask what shape an object would cast under certain rotations or lighting. For example, a sphere always projects a circle, while a cube may project a square or hexagon. Such reasoning strengthens geometric understanding.

Spatial Containment. Containment checks whether one object can fit inside another. For example, reasoning if a ball can fit in a box supports packing and design. This task aligns with intuitive geometry.

Gravity Effects. Gravity reasoning predicts motion outcomes under physical forces. For example, a rolling ball may stop under a table due to friction and barriers. This strengthens real-world simulation capabilities.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

We recruited annotators aged between 24 and 30 years, all of whom were engineering graduates. They participated voluntarily and worked independently without external incentives. Before starting the annotation process, we provided structured training sessions to ensure clarity of task requirements and consistency across annotators. We also maintained all ethical and quality control standards throughout the process. This approach ensured that the resulting annotations were both reliable and aligned with the dataset design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Before beginning annotation, we designed a structured training program to ensure consistency and reliability across the full taxonomy of tasks. We first introduced annotators to the six core categories (*Relative Positioning, Depth & Occlusion, Orientation, Size & Scale, Spatial Navigation, and 3D Geometry*) and their thirty subcategories. For each subcategory, we provided curated examples that illustrated the corresponding spatial relations (e.g., directional relations, occlusion inference, stability prediction) along with counter-examples to highlight common pitfalls. Beyond the taxonomy, we trained annotators to account for six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints, so that questions explicitly leveraged visual diversity. We demonstrated how these factors interact in real-world settings, such as reasoning about occlusion under low-contrast lighting or maintaining scale consistency under perspective distortion.

We also introduced the two evaluation modes: multiple-choice and open-ended. In multiple-choice settings, annotators learned to design precise distractors to test fine-grained discrimination, while in open-ended settings they practiced formulating prompts that encourage compositional reasoning. To ground this training, we ran calibration sessions in which annotators generated both types of questions for sample images and iteratively refined their designs with feedback. To situate task design within the broader research landscape, we reviewed prior benchmarks (e.g., SpatialBot-Bench, EmbSpatial, SpatialMM) with annotators and analyzed their limitations. These resources often emphasize simplified relations, synthetic or staged scenes, and narrow reasoning types, while neglecting natural complexity such as shadows, reflections, or gravity-sensitive stability. This gap analysis underscored the need for richer, more ambiguous, and dynamically layered spatial contexts.

Finally, we provided detailed demonstrations of the annotation protocol. These demonstrations specified how to select images with sufficient spatial complexity, how to map questions unambiguously to subcategories, and how to avoid trivial formulations. We instructed annotators to prioritize tasks requiring multi-step reasoning, such as depth layering, relational chains, or cross-view scale inference, over binary labels. Through this systematic training and iterative feedback, we ensured that the final dataset balanced taxonomic coverage with real-world relevance.

1458	
1459	
1460	
1461	
1462	
1463	
1464	
1465	
1466	
1467	
1468	
1469	
1470	
1471	
1472	
1473	
1474	
1475	
1476	
1477	
1478	
1479	
1480	
1481	
1482	
1483	
1484	
1485	
1486	
1487	
1488	
1489	
1490	
1491	
1492	
1493	
1494	
1495	
1496	
1497	
1498	
1499	
1500	
1501	
1502	
1503	
1504	
1505	
1506	
1507	
1508	
1509	
1510	
1511	

SpatialLab-MCQ : Direct : Prompt for Answering Model

You are a spatial reasoning assistant. Given a multiple choice question about an image, Answer with a single number (1-4) corresponding to the option.
 Question:
 {question}
 Options:
 1. {opt1}
 2. {opt2}
 3. {opt3}
 4. {opt4}

SpatialLab-Open : Direct : Prompt for Answering Model

You are a spatial reasoning assistant.
 Answer the following question in 2-4 sentences about the given image.
 Question:
 {question}

SpatialLab-Open : Direct : Prompt for Judge Model

You are a strict evaluator.
 You will receive a question, the ground truth answer, and a model's answer.
 Return only 1 if the model's answer is correct, otherwise 0.
 Question:
 {question}
 Ground Truth Answer:
 {ground_truth}
 Model Answer:
 {model_answer}
 Output only a single digit: 0 or 1.

Figure 6: Prompts used for evaluation.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

The prompts used for the answer model are shown in Figure 6. We set `temperature` to 0 and `max_tokens` to 10, while keeping all other generation parameters at their default values.

D.2 OPEN-ENDED EVALUATION

The prompts used for both the answer model and the judge model are provided in Figure 6. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other parameters kept at their default values.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To evaluate *Gemini-2.5-Flash* as an automatic judge, we selected a balanced set of 240 samples covering all categories. Each sample included the evaluation setup (question, ground truth, and model output), which was scored independently by three human annotators. Annotators assigned binary scores: 1 for correct and 0 for incorrect. We compared the LLM’s judgments against the human annotations using multiple complementary agreement metrics, computed both overall and per category. Specifically, we employed **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined as follows.

Cohen’s Kappa. Cohen’s kappa (κ) measures pairwise agreement between two raters while correcting for chance agreement. For binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e is the expected agreement by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here, $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c . An $\kappa = 1$: perfect agreement; $\kappa = 0$: agreement equals chance; $\kappa < 0$: systematic disagreement) We computed Cohen’s kappa between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), as well as between the LLM and the majority vote of the annotators (κ_{majority}).

Accuracy with Majority Vote. We also measured raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i , and y_i^{maj} is the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance agreement; it simply reflects the proportion of identical labels.

Fleiss’ Kappa. To assess inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values indicate stronger inter-annotator reliability beyond chance.

Table 5: Agreement metrics between Gemini-2.5-Flash as an automatic judge and three human annotators across all categories. The table reports Cohen’s kappa with the majority vote and each annotator, raw accuracy against the majority, and Fleiss’ kappa among all annotators. Bold values highlight the highest agreement within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation The most informative comparison in Table 5 is judge-vs-human against human-vs-human: the LLM judge (Gemini-2.5-Flash) reaches a Cohen’s κ of 0.738 with the majority vote (0.681–0.795 against individual annotators), essentially matching the 0.774 Fleiss’ κ that the humans achieve among themselves. In other words, the judge disagrees with the human consensus no more than the humans disagree with each other. Raw accuracy against the majority is 0.880 overall. The per-category picture tracks task clarity: agreement is perfect for 3D Geometry ($\kappa = 1.000$, accuracy 1.000) and high for Depth & Occlusion ($\kappa_{\text{majority}} = 0.805$, accuracy 0.909) and Spatial Navigation ($\kappa = 0.815$, accuracy 0.933), while the more subjective Orientation ($\kappa = 0.500$, accuracy 0.810) and Size & Scale ($\kappa = 0.417$, accuracy 0.714) sit lower but still within acceptable reliability.

Read together, these numbers place the judge’s agreement mostly in the substantial band (0.61–0.80) and, for clearly defined categories, into the almost-perfect band (> 0.81)—comparable to human evaluators throughout. The 0.774 inter-annotator Fleiss’ κ confirms the reference scores are themselves highly consistent, and the lower Orientation/Size & Scale figures reflect genuine task subjectivity rather than judge error. The automatic judge is therefore a faithful, quantitatively validated stand-in for human grading of the open-ended results.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) set the highest ceilings in the benchmark. GPT-4o-mini reaches 85.71% in *Stacking Orientation*—the top score overall—while Gemini-2.0-Flash leads on occlusion and projection (64.0% *Partial Occlusion*, 64.71% *Shadow-Size Projection*) and GPT-5-mini on embodied and layering cues (68.0% *Tool Handedness*, 55.71% *Layering Order*). The group’s floor, however, is just as telling: Claude 3.5 Haiku manages only 32.0% in *Relative Size Comparison*. Head to head, closed-source systems beat open-source ones by 15–25 points in several categories, yet their within-model spread still exceeds 50 points. The comparison is thus clear-cut: proprietary pipelines buy higher peaks but not the uniform spatial competence the benchmark demands.

Table 6: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

Open-source Large Models. Scaling to large open-source models (Table 7) raises the peaks without lifting the floor. InternVL3.5-72B reaches 80.56% in *Corner/Angle Positioning*—among the best open-source results—and both it and llama-3.2-90b-vision-instruct hit 66.67% in *Complete Occlusion Inference*; llama-3.2-90b also scores 71.43% in *Stacking Orientation*. Yet in the same group llama-3.2-11b-vision-instruct collapses to 2.0% in *Object Rotation*, gravity reasoning caps at a modest 56.0%, and the within-model spread often exceeds 70 points—wider than the smaller-model groups. The quantitative verdict is that size amplifies selective strengths while leaving the foundational representation gaps untouched.

Open-source Small Models. We observe that open-source small models (Table 8) display sharp contrasts in performance across sub-categories. For example, **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, while **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. Scores in *Pathway Existence* remain modest, with the best models tied at 34.78%. Gravity-related reasoning is similarly weak, with only 50.0% achieved by **InternVL3.5-4B** and **Gemma-3-4B-it**. The within-model spread often exceeds 60 percentage points, showing brittle generalization.

Table 8: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

These patterns suggest that smaller models are highly cue-dependent, performing well where visual templates are strong but failing in tasks requiring deeper spatial abstractions. We interpret this inconsistency as evidence that representation learning is uneven across sub-categories. Overall, we find that small models provide localized competence but lack robustness across the full spectrum of spatial reasoning.

Reasoning Models. Reasoning-enhanced models (Table 9) show notable benefits in select tasks but remain inconsistent overall. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, outperforming most open baselines. **Gemini-2.0-Flash-Thinking** achieves 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*, demonstrating stronger geometric reasoning. The strongest results come from **Gemini-2.5-Flash-Thinking**, which scores 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. Yet the same family fails in others, such as 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This variance shows that reasoning mechanisms improve depth in certain sub-categories but do not broaden coverage. Within-model spreads frequently exceed 35 percentage points. We conclude that reasoning augmentation adds localized capacity but leaves structural gaps in spatial understanding. Overall, these models demonstrate promise but not generalizable competence.

Spatial Reasoning Models. Spatially specialized models (Table 10) reveal targeted improvements but lack generality. **SpaceQwen2.5-VL-3B-Instruct** achieves 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, clearly surpassing many open-source baselines in size-scale reasoning. The same model leads in navigation tasks, with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*, matching expectations for domain-specific reasoning. However, performance drops sharply in other tasks, such as only 24.0% for *Spatial Containment*. Rarely do these models exceed 55.0%, even in their strongest areas. The spread across sub-categories frequently surpasses 30 percentage points. We find that specialization drives narrow peaks but does not produce holistic competence. Our interpretation is that domain targeting enhances select abilities but fails to integrate them into a robust, general framework.

Overall Discussion. Reading the five model groups side by side (Table 8, Table 7, Table 6, Table 10, Table 9), the same fragmentation recurs in every group, only at different heights. Small open-source models span 74.29% (*Stacking Orientation*) down to 12.0% (*Cardinal Direction*); larger models lift the ceiling to 80.56% (*Corner/Angle Positioning*) yet still bottom out at 2.0% (*Object Rotation*); closed-source systems set the overall peak at 85.71% (GPT-4o-mini, *Stacking Orientation*) but drop to 32.0% (*Relative Size Comparison*). Specialists top out below 55% even where they

Table 9: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Table 10: Multiple-choice Evaluation Accuracy on SPATIALAB-MCQ by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

are strongest (navigation 53.66%, stability 50.0%), and reasoning models pair highs like 71.79% (*Complete Occlusion Inference*) with lows near 34.0%. The common denominator is a within-model spread that regularly exceeds 50 percentage points—brittle internal consistency across every group. Sub-categories requiring physical abstraction, *Gravity Effects*, *Stability Prediction*, *Spatial Containment*, consistently score below 35–40% regardless of category. We attribute these failures to

Table 11: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-MCQ. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

two root causes: the absence of stable encodings of orientation and physics, and an over-reliance on surface-level correlations. The wide variance, from 85.71% highs to 2.0% lows, suggests models are pattern-matching rather than reasoning robustly about space. Scaling, proprietary training, and reasoning augmentations each provide benefits but none produce general coverage. Instead, each approach yields isolated peaks that coexist with glaring blind spots. The collective picture is one of fragmented competence rather than integrated ability. We conclude that progress in spatial reasoning will require architectural changes aimed at embedding consistent geometric grounding and physics-aware mechanisms. In sum, our results show both remarkable selective gains and the persistence of fundamental gaps in holistic spatial intelligence.

E.2 MODEL FAILURE ANALYSIS

The distribution across categories and sub-categories in Table 11 reveals several notable patterns in model performance. Categories such as 3D Geometry and Size & Scale show relatively balanced distributions, yet both retain a significant proportion of failures where no model answered correctly, suggesting that even basic geometric invariances are not reliably internalized. In contrast, Depth & Occlusion exhibits a heavier tail in the mid-range (2–4 correct), pointing to partial but inconsistent reasoning about layered scenes. Orientation emerges as a weak category overall, with especially low totals in Stacking Orientation and Tool Handedness, which indicates that reasoning about subtle directional or embodied affordances is poorly captured. Relative Positioning presents a mixed picture: while some tasks like Proximity Gradients show moderate success, others such as Corner/Angle Positioning have many near-zero correct counts, reflecting brittleness in compositional reasoning over reference frames. Interestingly, Spatial Navigation shows comparatively stronger performance, particularly in Obstacle Avoidance and Viewpoint Visibility, but still has significant errors in Pathway Existence where relational chaining is needed. When aggregated, no category achieves uniformly high accuracy, and the Grand Total shows an even spread across 0–5 bins, suggesting that success is highly task-dependent rather than reflecting a general spatial competence. Overall, models display isolated strengths but lack systematic reliability across spatial dimensions.

These results suggest that current vision–language models struggle with several root causes tied to representational and reasoning bottlenecks. High error rates in Stacking Orientation and Tool Handedness indicate limited grounding in physics-based and embodied reasoning, consistent with models trained primarily on static internet imagery rather than interactive or task-driven data. The difficulty in Corner/Angle Positioning and Pathway Existence reflects inadequate multi-step relational chaining, pointing to weaknesses in their ability to handle recursive spatial logic. The inconsistency in Transparency Effects and Reflective Surfaces underscores the challenge of handling visual phenomena that require non-local cues, such as secondary reflections or occluded geometry. By contrast, comparatively better performance in Obstacle Avoidance suggests that models may have learned shortcuts from dataset biases, such as frequent co-occurrence patterns of walkable paths, rather than genuine planning ability. Taken together, these failures highlight that model training regimes overemphasize correlation-driven visual-language alignments while underrepresenting embodied, counterfactual, and physics-based reasoning. Addressing these issues likely requires richer training distributions (e.g., embodied simulation data, physics-augmented learning) and architectural innovations that can explicitly model reference frames and relational constraints. Without such advances, models will continue to show brittle competence that breaks down under compositional or physically grounded tasks.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models post the strongest open-ended scores, but the group’s internal range is wide. GPT-5-mini is the clear leader in **3D Geometry** and **Relative Positioning**, peaking at 58.14% in Directional Relations and 55.81% in Alignment Patterns. Within the same group, however, Gemini-2.5-Flash falls to 16.28% on Proximity Gradients and Claude 3.5 Haiku to 12.0% on Perspective Distortion—gaps of roughly 40 points from the leader’s peaks. Even the best proprietary models are therefore fragile on subtle, continuous visual cues. We believe these disparities stem from the differential sophistication of models’ spatial reasoning pipelines, where some architectures capture relational and geometric structure more effectively than others. Tasks such as Proximity Gradients are particularly challenging, as they require analogical, graded reasoning rather than discrete classification. The fact that GPT-5-mini succeeds where others fail suggests that scaling and proprietary training corpora confer distinct advantages. Nevertheless, even the leader shows vulnerabilities, pointing to a systemic issue: closed-source dominance is relative, not absolute, and the models still fall short of robust, human-like spatial reasoning.

Open-source Large Models (Table 13). Large open-source models spread their strengths across different sub-categories rather than one system dominating. llama-3.2-90b-vision-instruct reaches 60.0% in Depth and Occlusion and GLM-4.5V-106B-MoE leads Spatial Containment at 38.0%. But the group shares a common failure point: qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct

Table 12: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Closed-source Models. We bold and underline the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Large Models. We bold and underline the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	16.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	24.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

both bottom out at 9.3% on Proximity Gradients—more than 50 points below the group’s occlusion peak. Scaling clearly helps peaks without guaranteeing coverage on the hardest continuous-reasoning tasks. We hypothesize that much of this variation arises from heterogeneous training data: models exposed to richer depth and geometry corpora perform better in occlusion and containment tasks, while others generalize poorly. The persistent weakness on Proximity Gradients suggests that these systems are not optimized for subtle spatial analogies and distance scaling. We also observe higher inter-model variance in this group than among closed-source models, underscoring the impact of open-source development heterogeneity. Overall, large open-source models show encouraging peaks but inconsistent coverage, suggesting that scale alone is insufficient without targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models predictably lag behind their larger counterparts, but their results reveal noteworthy localized strengths. qwen-2.5-vl-3b-instruct

Table 14: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Open-source Small Models. We bold and underline the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

achieves 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads in Spatial Containment (24.0%) and Stability Prediction (21.43%). Yet these successes are offset by catastrophic failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B manages only 2.0% in Shape Projection. We attribute this extreme variability to limited capacity and restricted training diversity, which prevent small models from learning robust representations of geometric and relational principles. Sub-categories that demand precise reasoning about orientation and hierarchy trigger complete breakdowns, as small misinterpretations translate into total failure. Nevertheless, their ability to occasionally outperform larger models in narrow tasks suggests that even small-scale training can instill isolated competencies. This points to the value of specialization but also confirms the structural ceiling of compact architectures. In short, small open-source models demonstrate pockets of promise but remain highly brittle, with overall accuracy limited by size and data constraints.

Reasoning Models (Table 15). Reasoning-oriented models present a sharper internal divide, with Gemini-2.5-flash-thinking standing far above its peers. It records the best scores in nearly every sub-category, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. By contrast, Kimi-VL-A3B-Thinking-2506 struggles dramatically, scoring just 6.0% in Spatial Containment and lagging behind across the board. We believe this gap arises from differences in how these models integrate multi-modal streams into coherent reasoning frameworks. Gemini-2.5-flash-thinking appears capable of forming and querying internal world models, enabling it to infer object interactions and physical properties more effectively. In contrast, weaker models rely heavily on surface-level pattern matching, which proves inadequate for compositional or causal reasoning. The consistently high performance of Gemini-2.5 across categories suggests that explicit reasoning mechanisms offer tangible benefits for spatial understanding. However, its failure to dominate Shape Projection and Betweenness Relationships shows that even advanced reasoning modules are not yet universally effective. Thus, while reasoning models hold promise, the category itself remains heterogeneous and far from solved.

Spatial Reasoning Models (Table 16). Spatial reasoning models, though explicitly designed for the task, record the lowest results overall. Performance is highly fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct stands out slightly with 23.91% in Volume Comparison and 13.73% in Scale Consistency, but these scores remain far below competitive thresholds. At the other extreme, Betweenness Relationships and Corner/Angle Positioning collapse entirely, with models recording 0.0% and 2.78%, respectively. These results suggest that the models’ smaller capacity and narrowly focused training severely constrain their representational flexibility. We interpret the failures as evidence that

Table 15: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended Evaluation Accuracy on SPATIALAB-OPEN by Question Sub-Categories for Spatial Reasoning Models. We bold and underline the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

narrow specialization without sufficient capacity or data breadth cannot yield generalizable spatial reasoning. Particularly on geometric and positional tasks, even slight representational errors lead to total breakdowns. Consequently, despite their focus, spatial reasoning models are least effective, underscoring the necessity of both scale and diversity in achieving robust performance.

Overall Discussion. Comparing the five groups side by side (Table 12, Table 13, Table 14, Table 15, Table 16) shows the same fragmented-competence signature at every level of the hierarchy. Closed-source models lead overall—GPT-5-mini peaks at 58.14% (Directional Relations) and 55.81% (Alignment Patterns)—yet still collapse on Proximity Gradients. Large open-source models reach 60.0% in Depth and Occlusion but 9.3% on Proximity Gradients, a 50-point intra-group swing that shows scale without specialization is insufficient. Small open-source models repeat the pattern at lower altitude, pairing 26.0% (Relative Size Comparison) with 0.0% (Betweenness Relationships). Reasoning models are the most internally divided: Gemini-2.5-flash-thinking hits 75.0% (Tool Handedness) while peers like Kimi-VL-A3B fall to single digits on the same tasks. The specialists, despite their focus, rank last overall—frequently below 15% and at 0.0% on Betweenness Relationships. The ordering (closed-source > large open > reasoning $\approx$ small open > specialists) is stable, and the recurring within-model swings above 50 points point to three root causes: (1) a lack of robust representations for continuous and analogical spatial cues, (2) structural limitations tied to scale and training diversity, and (3) brittle reasoning pipelines that prevent models from generalizing across tasks. We conclude that no current approach, whether proprietary scaling, open-source expansion, reasoning integration, or specialization, delivers comprehensive coverage. The frequent performance gaps exceeding 50 percentage points within a single model further confirm that progress is piecemeal. Future work must thus focus on unifying robust geometric encodings, physics-informed reasoning, and cross-task transfer, moving beyond isolated gains to achieve consistent spatial intelligence.

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 highlights persistent weaknesses across categories, with a notable skew toward the lower bins (0–2 models correct). 3D Geometry is broadly challenging, with Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) showing especially poor outcomes, suggesting models lack reliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, accumulating the highest zero-correct counts (72), particularly in Layering Order (20 zero) and Transparency Effects (16 zero), both requiring nuanced handling of occlusion and non-local cues. Orientation exhibits similar struggles: while Cardinal Direction achieves some mid-range success, Stacking Orientation (11 zero) and Tool Handedness (5 zero, none with perfect success) underscore the brittleness of embodied directional reasoning. Relative Positioning reveals widespread inconsistency, with tasks such as Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models, while Corner/Angle Positioning shows low but somewhat balanced performance. Size and Scale appears relatively stronger, with multiple subcategories (e.g., Scale Consistency with 6 full successes) showing that some models internalize coarse perspective and scaling cues, though failures remain high in Shadow-Size Projection (17 zero). Spatial Navigation is split: while Spatial Sequence and Obstacle Avoidance show occasional full-model success, Pathway Existence (16 zero) and Accessibility Constraints (10 zero) highlight persistent difficulty in multi-step relational chaining. Overall, the grand total reveals that across 356 items, zero-correct dominates (356), while only 74 items are solved by all five models, confirming that consistent spatial competence remains elusive.

These findings indicate that the root causes of failure stem less from dataset design and more from the representational and reasoning limitations of current models. First, the dominance of errors in Depth and Occlusion reflects a reliance on superficial appearance features rather than structured scene representations: for example, models may latch onto surface textures rather than reasoning about hidden layers, leading to systematic errors in Layering Order. Failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, since pretrained internet imagery often lacks explicit annotations of left/right handedness or stacking physics. The brittleness in Relative Positioning and Spatial Navigation points to deficiencies in chaining multi-hop spatial relations; for instance, Pathway Existence requires integrating local connectivity with global layout, which is not well captured by attention-based co-occurrence priors. On the other hand, partial success in Size and Scale suggests that models exploit statistical regularities in perspective distortion or scale consistency, though they collapse under less frequent cases like shadows. These patterns collectively show that current models succeed when cues are salient and correlate with training priors but fail when reasoning requires counterfactual, compositional, or embodied inference. Addressing these issues will require architectural advances that explicitly model reference frames, occlusion layers, and physical dynamics, as well as training data enriched with embodied simulations and counterfactual

Table 17: Distribution of question counts by the number of models (0–5) that correctly answered each item across categories and sub-categories in SPATIALAB-OPEN. “0” indicates no model answered correctly, while “5” indicates only five models succeeded to answer these among all models tested.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

augmentations. Without these, progress in spatial reasoning will remain fragmented, with models performing well on surface-level cues but breaking down on structured, real-world spatial logic.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This section quantifies, model by model and category by category, how much accuracy the shift from multiple-choice (MCQ) to open-ended answering costs on SPATIALAB, as reported in Table 18. The consistent, positive, and sizable gaps are the central evidence that a single MCQ number overstates practical spatial competence.

Table 18: **MCQ→open-ended accuracy gap (% , ↓) per spatial category on SPATIALAB**. Each cell is a model’s MCQ score minus its open-ended score, so smaller is better: a low value means the model performs comparably in both setups, while a large value flags accuracy that is inflated by the multiple-choice format. Across all 25 models the gap averages 23.0% ($\sigma = 5.5$); the per-group largest gap is bolded and underlined. Negative entries (e.g., -1.98 , -3.18) mark the rare cases where open-ended answering beats MCQ.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

G.1 QUANTITATIVE SNAPSHOT

Across the 25 models, the average MCQ→open-ended gap is **23.0%** ($\sigma = 5.5$), and it is strikingly uniform across categories: the per-subtask means fall within a 1.7-point band—spatial navigation 22.89, relative positioning 23.11, size & scale 23.42, depth & occlusion 23.45, orientation 23.70, and 3D geometry 24.57—for an overall 23.01. Individual overall gaps range from 11.93 to 31.57 (median ≈ 23.34). The ordering by model family is equally systematic: spatial reasoning models pay the largest penalty (27.03 on average), reasoning-oriented models the smallest (19.11), and proprietary models (21.09) fare better than the larger open-source models (24.51); for reference, even the human baseline shows a 22.64 gap. The extremes bracket the pattern—the widest gaps are InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%), and the narrowest are Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36%). Finally, correlation analysis identifies spatial navigation as the dominant driver of the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also strongly aligned.

G.2 PATTERNS AND OBSERVATIONS

Three broad patterns emerge, each with numbers attached. First, model family predicts robustness to format: specialist spatial reasoning models carry the largest gaps (27.03 average), consistent with optimization for classification-style supervision, whereas reasoning-focused models compress the gap to 19.11, reflecting the stabilizing effect of chain-of-thought-style training on generative outputs. Second, spatial navigation is the single strongest predictor of the overall disparity ($r = 0.99$): models that falter on open-ended navigation drive the aggregate gap, with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) contributing substantially and relative position and size/scale varying more idiosyncratically. Third, scale does not explain robustness—the smallest gap (Llama-3.2-11B, 11.93%) and the largest (InternVL3.5-72B, 31.57%) sit at opposite ends of the size spectrum. The

few negative gaps (depth & occlusion for Llama-3.2-11B at -1.98 ; relative position for o4-mini at -3.18) show that in isolated cases open-ended scoring is the more reliable of the two, likely because of misleading distractors.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize several interacting factors that underlie the observed MCQ→open-ended performance gaps in SPATIALAB, each supported by statistical observations:

- MCQ structural advantage.** MCQ format constrains the output space, allowing models to exploit surface cues or eliminate distractors. Across models, the mean MCQ score is systematically higher than the open-ended score, with the average gap at 23.0% and $\sigma = 5.5\%$. Negative gaps observed for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) indicate that open-ended generation can occasionally outperform MCQ when distractors are misleading, reinforcing the notion that MCQ can artificially inflate or distort measured performance.
Recommendation: To address this issue, we suggest complementing MCQ evaluations with open-ended tasks and subtask-level diagnostics, which can more accurately capture model competence and reduce overestimation caused by option-based cues.
- Specialization bias in spatial reasoning models.** Spatial reasoning models exhibit the largest average gaps (27.03%) despite being optimized for spatial tasks, suggesting that they may have been trained primarily on categorical or synthetic selection tasks rather than free-form generation. For instance, InternVL3.5-72B, a 72B parameter spatial reasoning model, shows the highest overall gap (31.57%), particularly in spatial navigation (36.68%) and orientation (34.44%), highlighting a reliance on classification-like supervision. Correlations between subtask gaps and overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) indicate that these format-dependent failures are concentrated in sequential and geometric reasoning tasks.
Recommendation: To mitigate this bias, we recommend fine-tuning or instruction-tuning on open-ended spatial generation tasks, which can improve model robustness to generative formats and reduce reliance on categorical supervision.
- Instruction-tuning and stepwise decoding.** Reasoning-oriented models (e.g., Gemini-2-Flash-Thinking, o4-mini) demonstrate smaller average gaps (around 19.11%) and lower variance across subtasks. For example, Gemini-2-Flash-Thinking has an overall gap of 13.0% and consistently modest subtask gaps (Difference in 3D Geometry: 13.9%, Difference Relative Positioning: 21.03%), suggesting that chain-of-thought or stepwise decoding helps models produce stable open-ended responses. This is further supported by reduced correlation between subtask gap variance and overall gap within this family, implying that instruction-tuned reasoning mitigates format sensitivity.
Recommendation: To address format sensitivity, we suggest adopting chain-of-thought or stepwise decoding strategies and instruction-tuning for open-ended spatial tasks, which can improve generative consistency and reduce MCQ dependence.
- Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models with high performance gaps in spatial navigation, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), also exhibit large overall gaps (29.78%), indicating that multi-step referential reasoning is disproportionately affected in open-ended formats. This suggests that MCQ options simplify the reasoning chain, masking underlying weaknesses in sequential grounding and relational computation.
Recommendation: To better capture sequential reasoning abilities, we recommend including dedicated open-ended spatio-navigation tasks and employing targeted multi-step reasoning supervision, which can improve grounding and reduce overestimation by MCQ.
- MCQ distractor artifacts.** Some negative or near-zero gaps indicate that poorly designed distractors can suppress measured MCQ performance. For example, o4-mini shows a negative relative position gap (-3.18%) while maintaining positive gaps in other subtasks, implying that distractor quality can artificially depress MCQ accuracy. Conversely, models like Qwen-VL2.5-32B-Instruct exhibit very high MCQ→open gaps (overall = 29.85%)

across multiple subtasks, suggesting that MCQ distractors can also exaggerate perceived competence in selection-based tasks.

Recommendation: To address this, we suggest careful auditing and refinement of MCQ distractors and complementing them with open-ended evaluation, which can yield more accurate and unbiased measures of model spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

The quantitative bottom line is that reporting both MCQ and open-ended results is not optional: relying on MCQ alone inflates estimates of practical spatial competence by about 23 points on average. Subtask breakdowns are equally essential, since spatial navigation ($r = 0.99$) disproportionately drives the observed differences and would be invisible in an aggregate score. Diagnostic evaluations that separate classification from generative reasoning isolate these error modes, and the family-level numbers suggest instruction-tuning and stepwise-decoding strategies—which cut the reasoning-model gap to 19.11—are the most promising levers for reducing MCQ dependence. Finally, the negative gaps make clear that careful design and auditing of MCQ distractors are necessary for fair cross-format comparison.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To measure how far spatial reasoning on our diverse benchmark can be improved, we evaluated four complementary strategies under a common protocol and report their gains and losses side by side.

H.1 INHERENT REASONING MECHANISMS

We isolate the effect of built-in reasoning by comparing each model against its own reasoning-enabled variant (Table 2, Table 3), holding everything else fixed. In MCQ, reasoning is a net positive across the board: Gemini-2.5-Flash-Thinking gains +4.64 points overall (52.93% vs. 48.29%), concentrated in Relative Positioning (+13.1) and Orientation (+8.6), and o4-mini-medium posts the highest overall MCQ accuracy of the reasoning group (53.21%). The pattern says explicit reasoning scaffolds pay off most in structured, discrete-answer settings.

Open-ended results are positive on average but more uneven. The reasoning variant again leads its baseline—Gemini-2.5-Flash-Thinking edges Gemini-2.5-Flash (32.77% vs. 30.93%)—while o4-mini-medium shows the largest margin at 37.86%, clearing every non-reasoning model. The stabilizing effect is strongest in complex categories such as 3D Geometry and Spatial Navigation, but it is not uniform: Gemini-2.5-Flash-Thinking regresses in Size & Scale (21.74% vs. 26.59%), a 4.85-point drop that echoes the format-driven brittleness seen elsewhere.

Taken together, the comparison quantifies a clear split: reasoning modules deliver consistent multi-point MCQ gains and modest open-ended stabilization, strengthening logical consistency and relational chaining, yet they leave scale sensitivity, occlusion, and free-form linguistic grounding unresolved.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To examine whether structured reasoning improves spatial performance, we further evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 sub-categories in SPATIALAB, we sampled 5 items (seed fixed to 42) to construct a balanced test set covering all major spatial phenomena. This design ensures fair coverage while keeping evaluation computationally tractable. Example CoT prompts are shown in Figure 7, which illustrate the step-by-step reasoning scaffolds provided to the models. The setup allows us to isolate the contribution of CoT reasoning from model scale or training.

The numbers in Table 19 are decisive: orientation is the only category where CoT reliably helps, while overall accuracy falls in three of the four model×format settings (−2.67 and −0.66 for InternVL-3-78b, −18.00 and −14.24 for Gemini-2.5-Flash). CoT thus helps when a task reduces to aligning directions but hurts when it demands perceptual priors (depth, scale, relative positioning). We attribute this to the absence of inherent 3D and geospatial knowledge: lacking human-like awareness

2268
2269
2270
2271
2272
2273
2274
2275
2276
2277
2278
2279
2280
2281
2282
2283
2284
2285
2286
2287
2288
2289
2290
2291
2292
2293
2294
2295
2296
2297
2298
2299
2300
2301
2302
2303
2304
2305
2306
2307
2308
2309
2310
2311
2312
2313
2314
2315
2316
2317
2318
2319
2320
2321

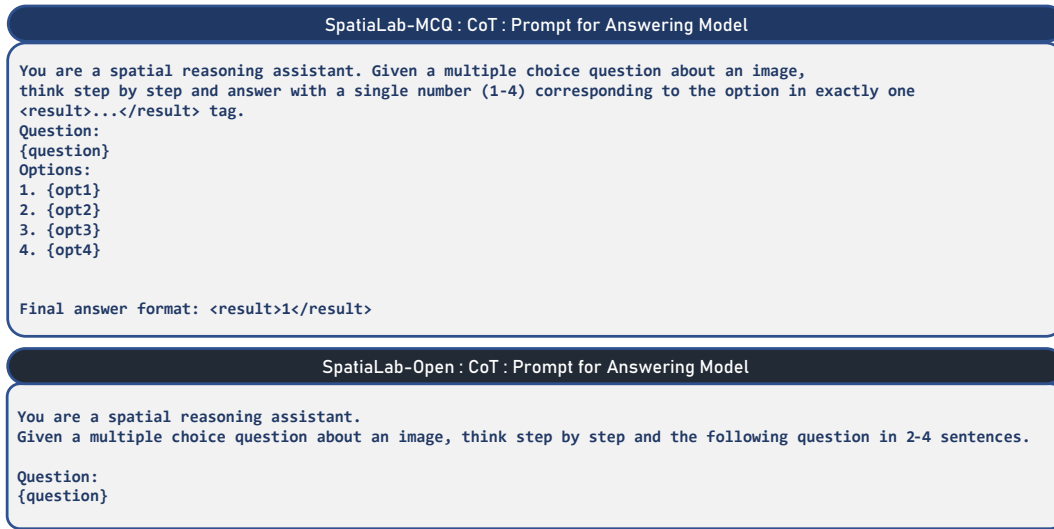


Figure 7: Chain-of-thought (CoT) Prompts.

Table 19: Effect of CoT prompting by category for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); Gain is the with-CoT minus without-CoT difference, so negative entries indicate CoT hurts. The overall Gains are negative in three of four settings (−2.67, −0.66, −18.00, −14.24), and Orientation is the only category with consistent positive gains—CoT rarely helps and often lowers accuracy on SPATIALAB.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

of these concepts, the model’s step-by-step chain does not correct misconceptions but propagates and amplifies them.

Quantitatively, this is what makes SPATIALAB non-trivial: adding reasoning steps cannot recover the gap, and in depth and size every extra step tends to move the model further from the answer. The most striking evidence is Gemini’s −18.00 / −14.24 overall MCQ/open drops under CoT—a reversal of the usual textual-reasoning trend—showing how fragile reasoning becomes without robust perceptual grounding.

Thus, our task design is a strength, it goes beyond surface-level reasoning and probes whether models possess a genuine understanding of geospatial and 3D relational concepts. Solving it requires more than chaining logic; it demands embeddings and training that capture core aspects of perception and spatial cognition. In this way, our benchmark challenges models at a fundamental level and highlights an important gap in current architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We further extend the CoT setup by adding a self-reflection stage, where the model is prompted to review and revise its initial reasoning before finalizing an answer (see Figure 8). The evaluation protocol remains identical to the standard CoT setting, using 5 balanced samples per sub-category (seed 42). This design isolates the effect of self-reflection while keeping the comparison directly aligned with baseline CoT performance.

Adding a self-reflection stage to CoT produces sharply model-dependent numbers. For InternVL-3-78b, overall accuracy declines (−2.66 MCQ, −8.67 open-ended), with the heaviest losses in depth,

2322
2323
2324
2325
2326
2327
2328
2329
2330
2331
2332
2333
2334
2335
2336
2337
2338
2339
2340
2341
2342
2343
2344
2345
2346
2347
2348
2349
2350
2351
2352
2353
2354
2355
2356
2357
2358
2359
2360
2361
2362
2363
2364
2365
2366
2367
2368
2369
2370
2371
2372
2373
2374
2375

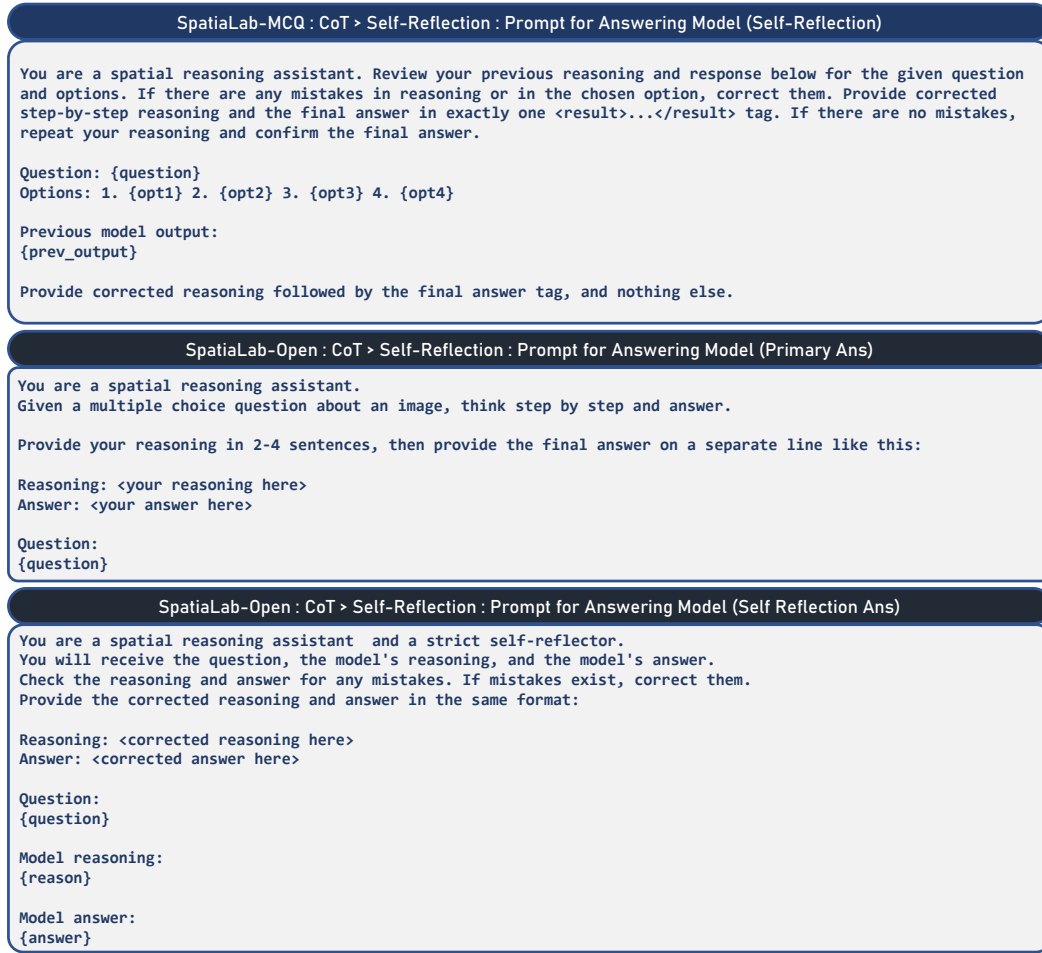


Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

Table 20: CoT vs. CoT + Self-Reflection (CoT-SR) by category for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); Gain is CoT-SR minus CoT. Self-reflection pays off mainly in the MCQ setting for Gemini-2.5-Flash (overall +20.66), but leaves InternVL-3-78b flat-to-negative (−2.66 MCQ, −8.67 open) and barely moves either model’s open-ended score (+0.67 for Gemini-2.5-Flash), confirming it cannot substitute for perceptual grounding.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

relative positioning, and navigation—reflection reinforcing rather than repairing errors; a localized +12 in Size & Scale (MCQ) is more than offset by a −28 collapse in open-ended relative positioning. Gemini-2.5-Flash moves the opposite way in MCQ, gaining +20.66 overall (with +32 in 3D Geometry and +24 in Depth & Occlusion), yet its open-ended change is a negligible +0.67. The asymmetry is the key quantitative finding: reflection helps precisely where option-pruning is possible (MCQ) and vanishes in free-form generation, where uncertainty propagates into ungrounded rationales. The gains also cluster in geometry-centric categories (orientation, depth, size) while multi-object relational

tasks (relative positioning, navigation) degrade—so on balance reflection stabilizes structured choice but does not reliably improve naturalistic reasoning.

This observed inconsistency mirrors our earlier findings with plain CoT prompting: reasoning enhancements only succeed when the model has stable perceptual anchors to ground its steps. Self-reflection adds an explicit verification pass, but this mechanism is only as strong as the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This explains why Gemini, which exhibits stronger perceptual grounding, benefits under MCQ conditions, while InternVL-3-78b suffers more frequent degradations. The broader root cause lies in a pretraining mismatch: current VLMs import reasoning templates from language modeling but lack the embedded 3D and geospatial representations that humans leverage. Without these anchors, reflection cannot perform true error correction; instead, it functions as a linguistic filter that polishes reasoning syntax without improving perceptual grounding. As a result, reflective reasoning loops provide modest utility in discrete answer formats but collapse under open-ended demands where factual verification is absent.

These findings confirm that our benchmark shows a fundamental gap in current architectures: surface-level reasoning layers alone cannot compensate for missing spatial cognition, and meaningful progress will require deeper integration of geometric and perceptual structure. Moreover, because VLLMs are commonly trained on web data, the concepts that are well-represented online are those where models tend to perform better, whereas poorly represented spatial and 3D concepts suffer, reinforcing these performance disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using a parameter-efficient approach with LoRA (Hu et al., 2021). To accommodate limited GPU resources, we loaded the model in 4-bit precision (load_in_4bit=True) and inserted LoRA adapters into attention and feed-forward network projection layers. The LoRA hyperparameters, including rank, scaling factor, dropout, and target modules, are summarized in Table 21. Gradient checkpointing was enabled to reduce memory usage, and a fixed random seed ensured reproducibility.

For tokenization, we followed the Qwen-2.5 chat template to maintain consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied with DataCollatorForSeq2Seq. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). Trainer and optimization hyperparameters, including learning rate, optimizer, weight decay, and scheduler, are listed in Table 22. The training loop consisted of four passes, each spanning a single epoch, yielding four saved checkpoints.

Overall, this setup allowed us to efficiently fine-tune a large vision-language model on limited GPU resources while maintaining high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a draw a clean contrast between the two evaluation modes. MCQ accuracy starts low at 25.63% and climbs monotonically to 36.59% by epoch four (+10.97 points), a steady curve indicating that fine-tuning reliably helps when answers are confined to a small candidate set. Open-ended accuracy behaves nothing like this: it starts higher at 34.40%, collapses to 12.62%

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

by the second epoch, and only claws back to 35.48% by the end—a net change of just +1.08 points despite the large mid-training swing. The MCQ curve therefore rises well above its starting point while the open-ended curve merely returns to it, and the gap between the two trajectories is the clearest evidence that SFT favors constrained-format stability.

Table 4b makes the split precise per category. In MCQ, every one of the six categories improves, from +1.59 in Relative Positioning to +17.11 in Size and Scale, for a grand-total gain of +10.97. The open-ended column is far more mixed: 3D Geometry (+3.50) and Spatial Navigation (+2.82) inch up, but Size and Scale drops −3.95 and Relative Positioning −0.79, leaving an overall open-ended gain of only +1.07—roughly one-tenth of the MCQ gain. The ten-fold difference between the two totals quantifies exactly how sensitive SFT is to the evaluation paradigm, and the per-category view pinpoints where training helps (structured discrimination) versus where it fails (free-form generation).

A category-by-category read exposes striking format disparities. In MCQ, Size and Scale (+17.11) and Spatial Navigation (+16.08) are the biggest winners, so multiple-choice fine-tuning is especially good at scale discrimination and navigational reasoning. Yet Size and Scale reverses to −3.95 in open-ended, a swing of over 21 points, showing the learned representations do not survive the move to generation. Relative Positioning tells the same story from the low end—barely +1.59 in MCQ and −0.79 in open-ended—a consistent relational weakness across formats. Orientation is the most robust category, pairing a strong +13.33 MCQ gain with a small positive open-ended change, the only category that generalizes cleanly. We hypothesize that the divergence between the two formats may stem from the structural scaffolding MCQs provide, which reduces the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, require simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends indicate that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and even degradation when free expression is required. This duality makes clear that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we show that SFT as currently applied provides strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not only an artifact of fine-tuning, but also symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image–text alignment, object recognition, and captioning, yet they provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. As a result, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, point to missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. Another contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions,

preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, we argue that without targeted reforms, current VLMs risk reinforcing an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To investigate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a distinct divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. Averaged over the five seeds (Table 23, Figure 9), SPATIALAB-MCQ improves monotonically from a 25.63% baseline to 35.74% at Epoch 4—a reproducible +10.11-point gain that confirms the model reliably adapts to the discriminative task.

SPATIALAB-OPEN follows the opposite, and equally reproducible, pattern: a pronounced *U-shape* across all five seeds. It collapses from 34.40% to an average of 19.26% at Epoch 1 (a −15.14-point drop), then recovers to ≈34.82% by Epoch 4, for a net change of just +0.42 over the zero-shot baseline. The two trajectories—+10.11 vs. +0.42—hold across every seed, so the format divergence is a stable property of SFT rather than a seed artifact.

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

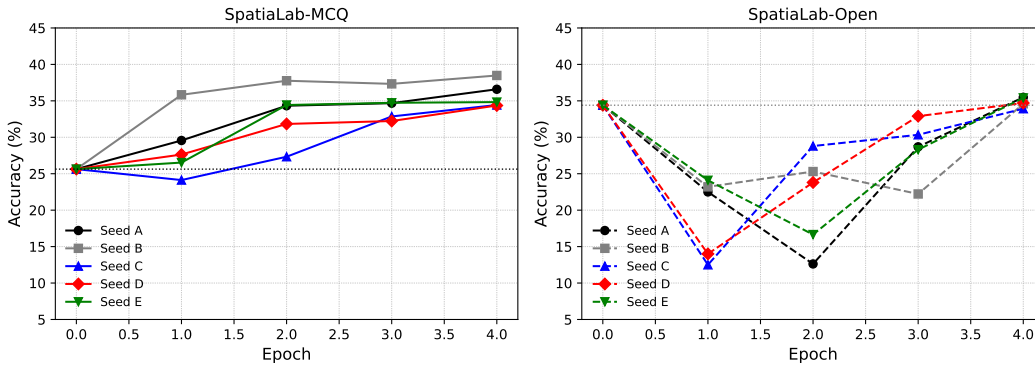


Figure 9: SFT Dynamics Analysis.

Representational Brittleness and Alignment Tax. We attribute this phenomenon to the fragility of spatial representations in current VLMs when subjected to fine-tuning. The initial collapse in open-ended accuracy suggests *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding required for open-ended descriptions.

The subsequent recovery phase represents the model re-learning to articulate spatial concepts within the new distribution, yet it fails to surpass the initial baseline significantly. This dynamic underscores a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not fundamentally enhance the underlying *generative spatial representation* needed for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To assess whether the representations learned from SPATIALAB generalize beyond the benchmark’s specific format, we evaluated the transferability of our fine-tuned model to three external spatial reasoning datasets: OMNISpatial, SPACE (Multimodal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

Table 24 shows that fine-tuning on SPATIALAB transfers, improving all three external benchmarks by a consistent +5.24 to +11.70 points in zero-shot. The largest gain is on MIND THE GAP (35.86% → 47.56%, +11.70), consistent with our benchmark’s emphasis on occlusion and spatial continuity, followed by OMNISpatial (40.30% → 47.35%, +7.05) and SPACE (23.43% → 28.67%, +5.24). That every target improves—none regresses—is direct evidence that the taxonomy teaches geometric and relational concepts aligned with broader definitions of spatial intelligence. These results validate the quality of the SPATIALAB data. While the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely encourage overfitting to a specific question template. Instead, the diversity of the 30 sub-categories forces the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIOLAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each managed by a dedicated agent, allowing for focused processing of objects, attributes, relations, symmetries, and transformations. This modular design not only improves accuracy and interpretability but also supports multi-step reasoning, enabling the system to propagate low-level perceptual cues through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are detailed in the following sections, with the full set of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including

geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing `objects`, `relations`, `groups`, `symmetries`, and `transformations`. Prompts enforce consistent object IDs and structured

formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For our technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to ensure deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition allows each specialized model to focus on a constrained sub-task, enhancing overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol remains identical to the standard CoT and CoT-SR setting, using 5 balanced samples per sub-category (seed 42), making a total of 150 samples.

Table 25: Agent Performance Analysis

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

The SPATIOXOLVER results are the most polarized of any intervention. Orientation is the sole category with consistent, large gains—+8.00 in MCQ and +36.00 in open evaluation (0.00%→36.00%), the single biggest improvement anywhere in our study—confirming that agentic decomposition excels when a task reduces to sequential alignment of directional cues.

Every other category tells the opposite story. Depth & occlusion suffers the largest loss, −24.00 in open evaluation (40.00%→16.00%), and spatial navigation drops −12.00; both demand 3D perceptual grounding that the priors cannot supply, so the extra reasoning steps entrench misconceptions. Relative positioning stagnates in MCQ and falls −14.00 in open evaluation, showing agent loops add no robustness where relational understanding is already weak.

The marginal positives do not survive the format change: size & scale gains +4.00 in MCQ but is flat-to-negative in open evaluation, and 3D geometry drops −4.00 (MCQ) and −12.00 (open). Netted out, agents move the overall score +2.66 in MCQ but −4.50 in open evaluation—one substantial win against a field of losses. The takeaway is quantitative: SPATIALAB cannot be solved by layering multi-step agentic reasoning on top of models that lack the geospatial and perceptual representations the tasks require.

Overall, the single-win-many-losses ledger (one category up, five flat-or-down in open evaluation) shows that agentic workflows alone cannot advance performance in this domain. Orientation benefits from structured reasoning, but the remaining categories expose a lack of inherent perceptual awareness that lets errors propagate and amplify through multi-step reasoning. SPATIALAB therefore acts as a strong stress test: the numbers say future progress will require embedding genuine perceptual and geospatial priors rather than relying on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section presents a focused qualitative analysis of model failures observed in our visual reasoning benchmark. We explain each error class, provide diagnostic root cause analysis, identify recurring patterns, and describe verification protocols. The discussion aims to be both interpretable and actionable, so it can inform future architectural and evaluation design.

I.1 OVERVIEW

Across the sample set, models achieve strong coarse perception, they can reliably recognize object categories, colors, and simple binary relations, but they break down systematically when multiple cues must be integrated. Failures are not random noise but cluster into a small set of interpretable classes: spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, which suggests that the underlying causes are common inductive biases in current vision-language pipelines rather than isolated training artifacts. Importantly, the same error often appears in both large-scale proprietary systems and small open-source baselines, indicating structural weaknesses shared across the model family. In the following, we unpack each error class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When queries require distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. For example, if asked to point out “the leftmost red cube among three stacked cubes,” the model may incorrectly output the middle cube, influenced by lighting highlights or central positioning. This problem becomes particularly acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. In such cases, the model’s errors are not random: it tends to privilege objects that appear most salient due to size, brightness, or centrality, even when these cues contradict the explicit spatial instructions in the query. These consistent patterns highlight that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes suggest that models lack a robust object-centric representation, or what cognitive science would call an “index” that persists across reasoning steps. Instead of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, often reassigning attention to whichever object seems most visually striking at intermediate layers. This leads to reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks involving relative comparisons, like “the ball behind the second chair”, particularly fragile. In essence, the model operates more like a salience detector than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, tend to mix contextual information from neighboring regions. While this is beneficial for global context, it erases the fine-grained, object-specific identity needed to distinguish between similar candidates. Once pooled, features of two adjacent red cubes may be nearly indistinguishable in the latent space. Without an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are inconsistently handled. On simple comparisons like “is a car larger than a bicycle,” models typically succeed because such relationships align with strong statistical priors. However, when faced with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A common pattern is reliance on canonical object sizes: for example, predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Another failure occurs in unusual visualizations, such as toy miniatures or architectural models, where familiar proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** These behaviors indicate that models rely heavily on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works well in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. For instance, when an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors highlight the lack of grounding in pixel-level metric cues and the absence of consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures do not include explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but not the underlying geometry. Furthermore, training objectives prioritize overall accuracy rather than fine-grained metric consistency, meaning that scale-sensitive errors are under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes rather than interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models often misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. For instance, in an image where a cat sits partially behind a curtain, models may incorrectly describe the curtain as being occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are particularly error-prone: they are frequently omitted altogether, leading to wrong answers in questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or when the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. While the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them with sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, effectively removing the very cues necessary to resolve ordering. This is why models succeed in gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice, they guess rather than infer, with systematic biases toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and has no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are easily lost when features are pooled across large receptive fields. Furthermore, training data often underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries. As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models may base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Similarly, a surface with a metallic texture may be misclassified as “steel” even if it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. These confusions are

frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions when visual input is ambiguous. The decoder favors plausible narratives based on co-occurrence in the training data: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut allows models to produce semantically coherent answers even when they do not inspect the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes rather than grounding in perceptual details.

⊗ **Root cause.** The underlying training objective, maximizing likelihood of fluent and plausible text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized, even if the reasoning is incorrect. Consequently, the model is incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps to tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models often generate coherent but unsupported narratives. For example, when asked “why is the object leaning,” a model might answer “because the surface is slippery” even though slipperiness cannot be visually inferred. In another case, a tilted shelf might be explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors highlight a tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but there is no mechanism ensuring that these rationales are tied to actual visual features. Confidence calibration compounds the problem: models often assign high certainty to fabricated explanations, giving the false impression of reliability. As a result, open-ended outputs may look authoritative while being factually baseless. This mode of failure is particularly concerning in safety-critical domains where plausible but false rationales could misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, without constraints linking the explanation back to specific image evidence. The lack of an explicit verification layer or grounding requirement means that models default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints, open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

When comparing failures across different models, several consistent patterns appear, revealing insights into the underlying mechanisms of errors. One clear trend is that disagreements between models are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This leads to predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks that require reasoning about occlusion, relative scale, and perspective together are particularly challenging; for instance, understanding whether a larger object partially occluding a smaller one is in front or behind often triggers consistent mispredictions. Calibration is another noticeable issue: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns indicate that errors arise from how models balance perceptual grounding and linguistic plausibility. Different architectures exhibit distinct tendencies, some over-rely on visual geometry, others overfit to language correlations, so failures are not just isolated mistakes but reflect deeper trade-offs in reasoning strategies. This also implies that any attempt to improve performance should account for both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can be pursued to reduce failures without requiring complete architectural redesign. For instance, geometry-aware supervision can be incorporated by adding auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation. This encourages models to respect spatial constraints and improves consistency across predictions. Multi-scale feature retention is another approach: by using edge-aware pooling or attention mechanisms, models can maintain information from thin or partially occluded structures that would otherwise vanish in standard downsampling. Coordinate anchoring can help resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Counterfactual augmentation can also be applied, where distractor objects are systematically swapped or perturbed, forcing the model to rely on the true referent cues rather than heuristics. Lightweight verification modules can be added on top of existing outputs, cross-checking claims against visual or geometric constraints to filter implausible predictions. Each of these strategies can be implemented incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference resolution traps, SFT can directly expose models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skipping to a guess. This not only strengthens grounding but also improves interpretability, as errors become traceable to specific reasoning steps. Moreover, SFT can be paired with counterfactual or adversarial augmentations, ensuring that the model does not simply memorize but instead learns invariances to distractors and context shifts. Importantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this sense, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and observed clear performance gains, particularly on multiple-choice questions. This suggests that targeted supervision helps models better internalize spatial cues and resolve distractor confusions when answer options are well-defined. However, improvements on open-ended responses were more limited. The model continues to produce fluent but weakly grounded explanations, highlighting that SFT alone cannot fully address rationalization without verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve balanced progress across both structured and free-form tasks.

Our analysis shows that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. This indicates that VLM errors follow systematic trajectories tied to the inductive biases of their architectures and training regimes. For example, geometric reasoning often collapses when occlusion cues are weak, and in such cases language priors take over, producing answers that sound plausible but lack visual grounding. Similarly, scale inference drifts when canonical object size priors dominate over pixel-level metric cues, leading to consistent misjudgments in near-threshold scenarios. These findings highlight that the gap is not simply one of capacity but of representation: current models lack explicit mechanisms to anchor visual features to grounded geometric relations.

By adopting the proposed verification protocols and targeted mitigation strategies, researchers can address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation would help models resist shortcuts, while multi-scale feature retention ensures thin structures and partial occlusions are not discarded. SFT, when carefully applied, can further strengthen structured reasoning by aligning model predictions with task constraints, though it must be combined with grounding objectives to avoid overfitting to discrete-choice formats. Importantly, verification modules provide a lightweight way to enforce consistency between claimed relations and image evidence, counteracting the tendency to rationalize without checking. Taken together, these steps chart a practical pathway toward systems that reason more like humans: robust to clutter, interpretable in their failures, and capable of explaining not only what they predict but why.

2916
2917
2918
2919
2920
2921
2922
2923
2924
2925
2926
2927
2928
2929
2930
2931
2932
2933
2934
2935
2936
2937
2938
2939
2940
2941
2942
2943
2944
2945
2946
2947
2948
2949
2950
2951
2952
2953
2954
2955
2956
2957
2958
2959
2960
2961
2962
2963
2964
2965
2966
2967
2968
2969

Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position": "Leftmost" }},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position": "Center" }}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

2970
2971
2972
2973
2974
2975
2976
2977
2978
2979
2980
2981
2982
2983
2984
2985
2986
2987
2988
2989
2990
2991
2992
2993
2994
2995
2996
2997
2998
2999
3000
3001
3002
3003
3004
3005
3006
3007
3008
3009
3010
3011
3012
3013
3014
3015
3016
3017
3018
3019
3020
3021
3022
3023

Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

3024
3025
3026
3027
3028
3029
3030
3031
3032
3033
3034
3035
3036
3037
3038
3039
3040
3041
3042
3043
3044
3045
3046
3047
3048
3049
3050
3051
3052
3053
3054
3055
3056
3057
3058
3059
3060
3061
3062
3063
3064
3065
3066
3067
3068
3069
3070
3071
3072
3073
3074
3075
3076
3077

Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, illustrating the diverse spatial reasoning challenges covered. These samples highlight the variety of visual phenomena, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks. Together, they provide a concrete view of the types of reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

3132
3133
3134
3135
3136
3137
3138
3139
3140
3141
3142
3143
3144
3145
3146
3147
3148
3149
3150
3151
3152
3153
3154
3155
3156
3157
3158
3159
3160
3161
3162
3163
3164
3165
3166
3167
3168
3169
3170
3171
3172
3173
3174
3175
3176
3177
3178
3179
3180
3181
3182
3183
3184
3185

3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

3186
3187
3188
3189
3190
3191
3192
3193
3194
3195
3196
3197
3198
3199
3200
3201
3202
3203
3204
3205
3206
3207
3208
3209
3210
3211
3212
3213
3214
3215
3216
3217
3218
3219
3220
3221
3222
3223
3224
3225
3226
3227
3228
3229
3230
3231
3232
3233
3234
3235
3236
3237
3238
3239

3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

3240
3241
3242
3243
3244
3245
3246
3247
3248
3249
3250
3251
3252
3253
3254
3255
3256
3257
3258
3259
3260
3261
3262
3263
3264
3265
3266
3267
3268
3269
3270
3271
3272
3273
3274
3275
3276
3277
3278
3279
3280
3281
3282
3283
3284
3285
3286
3287
3288
3289
3290
3291
3292
3293

3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

3294
3295
3296
3297
3298
3299
3300
3301
3302
3303
3304
3305
3306
3307
3308
3309
3310
3311
3312
3313
3314
3315
3316
3317
3318
3319
3320
3321
3322
3323
3324
3325
3326
3327
3328
3329
3330
3331
3332
3333
3334
3335
3336
3337
3338
3339
3340
3341
3342
3343
3344
3345
3346
3347

3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

3348
3349
3350
3351
3352
3353
3354
3355
3356
3357
3358
3359
3360
3361
3362
3363
3364
3365
3366
3367
3368
3369
3370
3371
3372
3373
3374
3375
3376
3377
3378
3379
3380
3381
3382
3383
3384
3385
3386
3387
3388
3389
3390
3391
3392
3393
3394
3395
3396
3397
3398
3399
3400
3401

Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

3402
3403
3404
3405
3406
3407
3408
3409
3410
3411
3412
3413
3414
3415
3416
3417
3418
3419
3420
3421
3422
3423
3424
3425
3426
3427
3428
3429
3430
3431
3432
3433
3434
3435
3436
3437
3438
3439
3440
3441
3442
3443
3444
3445
3446
3447
3448
3449
3450
3451
3452
3453
3454
3455

Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

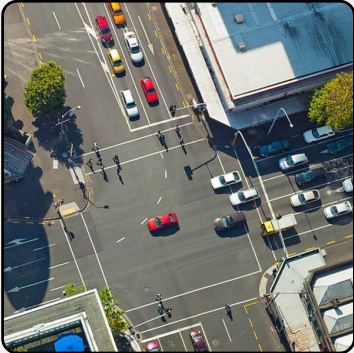
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

3456
3457
3458
3459
3460
3461
3462
3463
3464
3465
3466
3467
3468
3469
3470
3471
3472
3473
3474
3475
3476
3477
3478
3479
3480
3481
3482
3483
3484
3485
3486
3487
3488
3489
3490
3491
3492
3493
3494
3495
3496
3497
3498
3499
3500
3501
3502
3503
3504
3505
3506
3507
3508
3509

Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

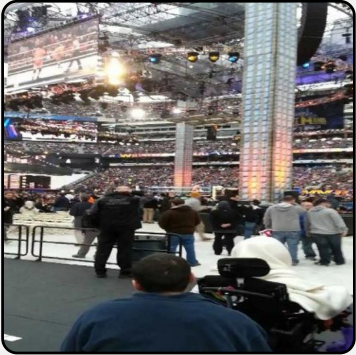
Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

3510
3511
3512
3513
3514
3515
3516
3517
3518
3519
3520
3521
3522
3523
3524
3525
3526
3527
3528
3529
3530
3531
3532
3533
3534
3535
3536
3537
3538
3539
3540
3541
3542
3543
3544
3545
3546
3547
3548
3549
3550
3551
3552
3553
3554
3555
3556
3557
3558
3559
3560
3561
3562
3563

Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

3564
3565
3566
3567
3568
3569
3570
3571
3572
3573
3574
3575
3576
3577
3578
3579
3580
3581
3582
3583
3584
3585
3586
3587
3588
3589
3590
3591
3592
3593
3594
3595
3596
3597
3598
3599
3600
3601
3602
3603
3604
3605
3606
3607
3608
3609
3610
3611
3612
3613
3614
3615
3616
3617

Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

3618
3619
3620
3621
3622
3623
3624
3625
3626
3627
3628
3629
3630
3631
3632
3633
3634
3635
3636
3637
3638
3639
3640
3641
3642
3643
3644
3645
3646
3647
3648
3649
3650
3651
3652
3653
3654
3655
3656
3657
3658
3659
3660
3661
3662
3663
3664
3665
3666
3667
3668
3669
3670
3671

Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

3672
3673
3674
3675
3676
3677
3678
3679
3680
3681
3682
3683
3684
3685
3686
3687
3688
3689
3690
3691
3692
3693
3694
3695
3696
3697
3698
3699
3700
3701
3702
3703
3704
3705
3706
3707
3708
3709
3710
3711
3712
3713
3714
3715
3716
3717
3718
3719
3720
3721
3722
3723
3724
3725

Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

3726
3727
3728
3729
3730
3731
3732
3733
3734
3735
3736
3737
3738
3739
3740
3741
3742
3743
3744
3745
3746
3747
3748
3749
3750
3751
3752
3753
3754
3755
3756
3757
3758
3759
3760
3761
3762
3763
3764
3765
3766
3767
3768
3769
3770
3771
3772
3773
3774
3775
3776
3777
3778
3779

Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

3780
3781
3782
3783
3784
3785
3786
3787
3788
3789
3790
3791
3792
3793
3794
3795
3796
3797
3798
3799
3800
3801
3802
3803
3804
3805
3806
3807
3808
3809
3810
3811
3812
3813
3814
3815
3816
3817
3818
3819
3820
3821
3822
3823
3824
3825
3826
3827
3828
3829
3830
3831
3832
3833

Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

3834
3835
3836
3837
3838
3839
3840
3841
3842
3843
3844
3845
3846
3847
3848
3849
3850
3851
3852
3853
3854
3855
3856
3857
3858
3859
3860
3861
3862
3863
3864
3865
3866
3867
3868
3869
3870
3871
3872
3873
3874
3875
3876
3877
3878
3879
3880
3881
3882
3883
3884
3885
3886
3887

Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

3888
3889
3890
3891
3892
3893
3894
3895
3896
3897
3898
3899
3900
3901
3902
3903
3904
3905
3906
3907
3908
3909
3910
3911
3912
3913
3914
3915
3916
3917
3918
3919
3920
3921
3922
3923
3924
3925
3926
3927
3928
3929
3930
3931
3932
3933
3934
3935
3936
3937
3938
3939
3940
3941

Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- **GPT-5-mini: 2**
- **Gemini-2.5-Flash: 2**
- **InternVL3.5-72B: 2**
- **Kimi-VL-A3B-Thinking-2506: 2**

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

3942
3943
3944
3945
3946
3947
3948
3949
3950
3951
3952
3953
3954
3955
3956
3957
3958
3959
3960
3961
3962
3963
3964
3965
3966
3967
3968
3969
3970
3971
3972
3973
3974
3975
3976
3977
3978
3979
3980
3981
3982
3983
3984
3985
3986
3987
3988
3989
3990
3991
3992
3993
3994
3995

Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

3996
3997
3998
3999
4000
4001
4002
4003
4004
4005
4006
4007
4008
4009
4010
4011
4012
4013
4014
4015
4016
4017
4018
4019
4020
4021
4022
4023
4024
4025
4026
4027
4028
4029
4030
4031
4032
4033
4034
4035
4036
4037
4038
4039
4040
4041
4042
4043
4044
4045
4046
4047
4048
4049

Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

4050
4051
4052
4053
4054
4055
4056
4057
4058
4059
4060
4061
4062
4063
4064
4065
4066
4067
4068
4069
4070
4071
4072
4073
4074
4075
4076
4077
4078
4079
4080
4081
4082
4083
4084
4085
4086
4087
4088
4089
4090
4091
4092
4093
4094
4095
4096
4097
4098
4099
4100
4101
4102
4103

Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

4104
4105
4106
4107
4108
4109
4110
4111
4112
4113
4114
4115
4116
4117
4118
4119
4120
4121
4122
4123
4124
4125
4126
4127
4128
4129
4130
4131
4132
4133
4134
4135
4136
4137
4138
4139
4140
4141
4142
4143
4144
4145
4146
4147
4148
4149
4150
4151
4152
4153
4154
4155
4156
4157

Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

1. 7
2. 6
3. 5
4. 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

4158
4159
4160
4161
4162
4163
4164
4165
4166
4167
4168
4169
4170
4171
4172
4173
4174
4175
4176
4177
4178
4179
4180
4181
4182
4183
4184
4185
4186
4187
4188
4189
4190
4191
4192
4193
4194
4195
4196
4197
4198
4199
4200
4201
4202
4203
4204
4205
4206
4207
4208
4209
4210
4211

Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

4212
4213
4214
4215
4216
4217
4218
4219
4220
4221
4222
4223
4224
4225
4226
4227
4228
4229
4230
4231
4232
4233
4234
4235
4236
4237
4238
4239
4240
4241
4242
4243
4244
4245
4246
4247
4248
4249
4250
4251
4252
4253
4254
4255
4256
4257
4258
4259
4260
4261
4262
4263
4264
4265

Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini:** 3
- Gemini-2.5-Flash:** 3
- InternVL3.5-72B:** 4
- Kimi-VL-A3B-Thinking-2506:** 4

Open-ended Answers:

- GPT-5-mini:** If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash:** The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B:** If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506:** The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

4266
4267
4268
4269
4270
4271
4272
4273
4274
4275
4276
4277
4278
4279
4280
4281
4282
4283
4284
4285
4286
4287
4288
4289
4290
4291
4292
4293
4294
4295
4296
4297
4298
4299
4300
4301
4302
4303
4304
4305
4306
4307
4308
4309
4310
4311
4312
4313
4314
4315
4316
4317
4318
4319

Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

4320
4321
4322
4323
4324
4325
4326
4327
4328
4329
4330
4331
4332
4333
4334
4335
4336
4337
4338
4339
4340
4341
4342
4343
4344
4345
4346
4347
4348
4349
4350
4351
4352
4353
4354
4355
4356
4357
4358
4359
4360
4361
4362
4363
4364
4365
4366
4367
4368
4369
4370
4371
4372
4373

Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

4374
4375
4376
4377
4378
4379
4380
4381
4382
4383
4384
4385
4386
4387
4388
4389
4390
4391
4392
4393
4394
4395
4396
4397
4398
4399
4400
4401
4402
4403
4404
4405
4406
4407
4408
4409
4410
4411
4412
4413
4414
4415
4416
4417
4418
4419
4420
4421
4422
4423
4424
4425
4426
4427

Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

4428
4429
4430
4431
4432
4433
4434
4435
4436
4437
4438
4439
4440
4441
4442
4443
4444
4445
4446
4447
4448
4449
4450
4451
4452
4453
4454
4455
4456
4457
4458
4459
4460
4461
4462
4463
4464
4465
4466
4467
4468
4469
4470
4471
4472
4473
4474
4475
4476
4477
4478
4479
4480
4481

Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

4482
4483
4484
4485
4486
4487
4488
4489
4490
4491
4492
4493
4494
4495
4496
4497
4498
4499
4500
4501
4502
4503
4504
4505
4506
4507
4508
4509
4510
4511
4512
4513
4514
4515
4516
4517
4518
4519
4520
4521
4522
4523
4524
4525
4526
4527
4528
4529
4530
4531
4532
4533
4534
4535

Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

4536
4537
4538
4539
4540
4541
4542
4543
4544
4545
4546
4547
4548
4549
4550
4551
4552
4553
4554
4555
4556
4557
4558
4559
4560
4561
4562
4563
4564
4565
4566
4567
4568
4569
4570
4571
4572
4573
4574
4575
4576
4577
4578
4579
4580
4581
4582
4583
4584
4585
4586
4587
4588
4589

Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

4590
4591
4592
4593
4594
4595
4596
4597
4598
4599
4600
4601
4602
4603
4604
4605
4606
4607
4608
4609
4610
4611
4612
4613
4614
4615
4616
4617
4618
4619
4620
4621
4622
4623
4624
4625
4626
4627
4628
4629
4630
4631
4632
4633
4634
4635
4636
4637
4638
4639
4640
4641
4642
4643

Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Safari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

4644
4645
4646
4647
4648
4649
4650
4651
4652
4653
4654
4655
4656
4657
4658
4659
4660
4661
4662
4663
4664
4665
4666
4667
4668
4669
4670
4671
4672
4673
4674
4675
4676
4677
4678
4679
4680
4681
4682
4683
4684
4685
4686
4687
4688
4689
4690
4691
4692
4693
4694
4695
4696
4697

Relative Positioning : Proximity Gradients

Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- **Gemini-2.5-Flash:** "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- **InternVL3.5-72B:** The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- **Kimi-VL-A3B-Thinking-2506:** Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

Because the paper’s central claims are quantitative comparisons—between models, formats, and subcategories—we first establish that the reported numbers are precise enough to support them. We therefore measure both model robustness across runs and dataset stability across repeated trials; such analyses are essential for small-to-medium benchmarks, where minor fluctuations can otherwise produce misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010). As the results below show, the measurement noise is an order of magnitude smaller than the accuracy gaps we report, so those gaps reflect capability, not chance.

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of the same model. Each run uses a distinct random seed while preserving all other hyperparameters.

K.1 MODEL ROBUSTNESS

For checking model evaluation robustness, we perform several statistical studies, as described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a}_i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify agreement between runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979), using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also analyze the stability of the dataset itself, whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further assess **robustness**, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test assesses median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we again fail to reject H_0 , confirming that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the correlation between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) suggests stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby covering both large- and small-scale models as well as proprietary and open-source paradigms. We also conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for `wilcoxon` test; `pingouin`³ (Vallat, 2018) for `intraclass correlation` and `cronbach alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), indicating that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Model Accuracy and Variance across 3 Independent Runs.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The run-to-run standard deviations are uniformly tiny—0.0021–0.0054 in MCQ and 0.0042–0.0047 in open-ended—i.e., all below 0.6%. Put in context, this is one to two orders of magnitude smaller than the multi-point and tens-of-point gaps the paper reports between models and formats, so those differences cannot be explained by seeding or decoding noise. Open-ended evaluation shows marginally higher variance than MCQ, but the absolute figures remain negligible, confirming that free-form generation is scored just as stably. Because estimates already converge within a single run, single-run numbers are reliable for large-scale comparison and the multi-run analysis chiefly certifies that robustness—strengthening the credibility of every cross-model and cross-format claim.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) Reliability Scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

Every ICC(3,k) value lands in the “excellent” band—0.988/0.990 in MCQ and 0.983/0.981 in open-ended, all above the 0.90 excellent-consistency threshold. This means the item-level difficulty ordering is essentially fixed across runs, so relative rankings among samples do not shift with stochastic variation and random guessing is ruled out even in the higher-variance open-ended setting. In practical terms, the benchmark yields a dependable capability signal, and the fine-grained cross-model and cross-format comparisons throughout the paper can be read with high confidence in measurement stability.

Taken together, the two axes agree: absolute performance is stable ($\sigma_A < 0.6\%$) and relative performance is near-perfectly preserved ($\text{ICC}(3,k) \geq 0.981$). Both being at ceiling reliability simultaneously

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

is the strongest possible guarantee that the performance gaps we report across models and formats are genuine capability differences rather than seeding or decoding noise, giving the evaluation framework a reliable and reproducible basis for comparison.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling Study (1,000 Trials): Wilcoxon Signed-Rank Test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

Across all 1,000 trials the Wilcoxon test never once detected a significant difference between the 20- and 25-sample subsets: mean p -values are 0.285–0.294 and 100% of trials satisfy $p > 0.05$ for every model and format. This uniform non-significance shows accuracy estimates have already converged at small subcategory sizes and are insensitive to moderate sample reductions, independent of model scale or output structure. The variance in accuracy is therefore dominated by model behavior, not data sparsity, and the current subcategory size provides a comfortable safety margin beyond what reliable estimation requires—evidence that the benchmark design is statistically sound and that future extensions could reallocate samples without sacrificing robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-Level Consistency Metrics.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

Mean pairwise correlations are uniformly high—0.966/0.972 (MCQ) and 0.950/0.946 (open-ended), all above 0.94—showing that the relative difficulty of individual questions is preserved run to run rather than fluctuating under stochastic decoding. The matching Cronbach’s α values (0.981–0.990, all above the 0.90 excellent threshold) confirm that items contribute coherently to the overall score. The slight dip in the open-ended figures is the expected cost of greater expressive freedom and still sits firmly in the excellent range. The benchmark thus behaves as a single, internally consistent instrument, so measured performance differences can be attributed to true capability variation rather than item-level instability.

K.3.3 OVERALL DISCUSSION

The robustness evidence converges on a single quantitative bottom line: **ICC and Cronbach’s α consistently exceed 0.98, run-to-run accuracy standard deviation stays below 0.6%, and the resampling study returns no significant difference in 100% of 1,000 trials.** Because this noise floor is far smaller than the accuracy gaps the paper reports, the granular comparisons across models, formats, and subcategories reflect genuine capability differences rather than statistical artifacts.

In sum, the evaluation protocol and dataset are highly robust. Near-perfect reliability and internal consistency (ICC and α above 0.98), negligible cross-run variance ($< 0.6\%$), and statistically

indistinguishable estimates from smaller subsets ($N = 20$ vs. $N = 25$) together establish that the chosen subcategory size sits well beyond the convergence threshold. Both absolute scores and relative rankings are stable across runs and sampling regimes, so every fine-grained comparison in the paper is meaningful and attributable to model capability rather than sampling variance or evaluation noise.